

Beam Diagnostics for JLAB

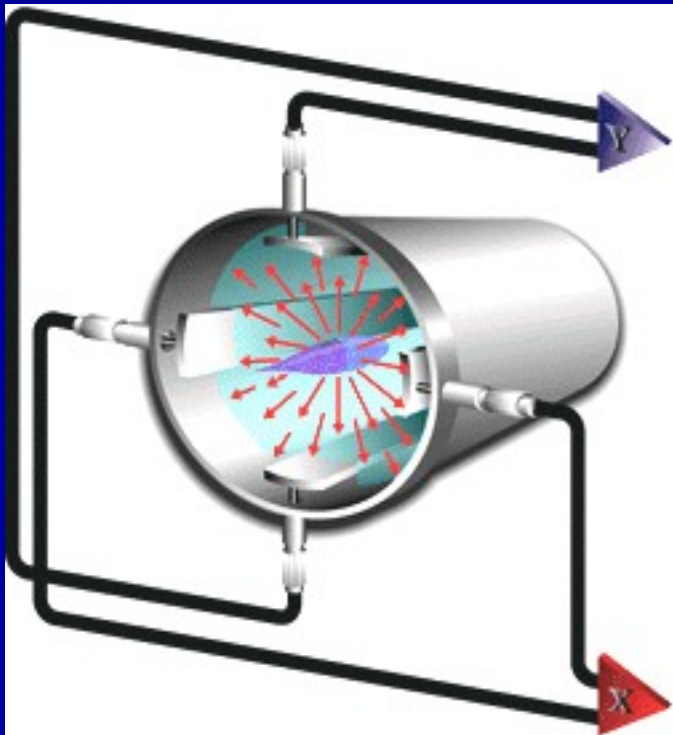
John Musson
JLAB I&C Group

Overview

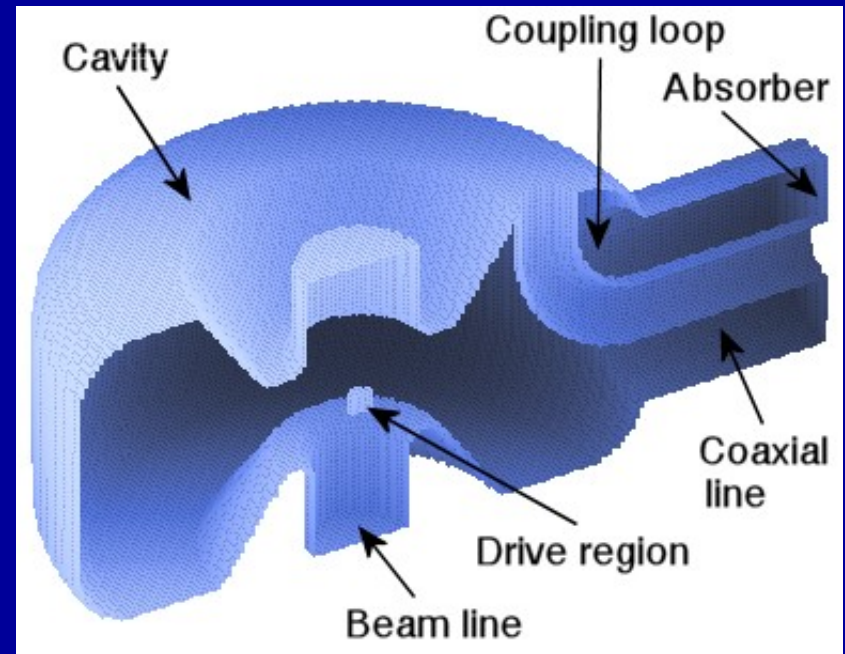
- Beamline Sensors
- Signals, Noise, and Input Parameters
 - Tutorial
- Receiver Electronics
 - Algorithms
- Bench Testing
- New M15
- Conclusions
- References

Typical Beamline Sensors

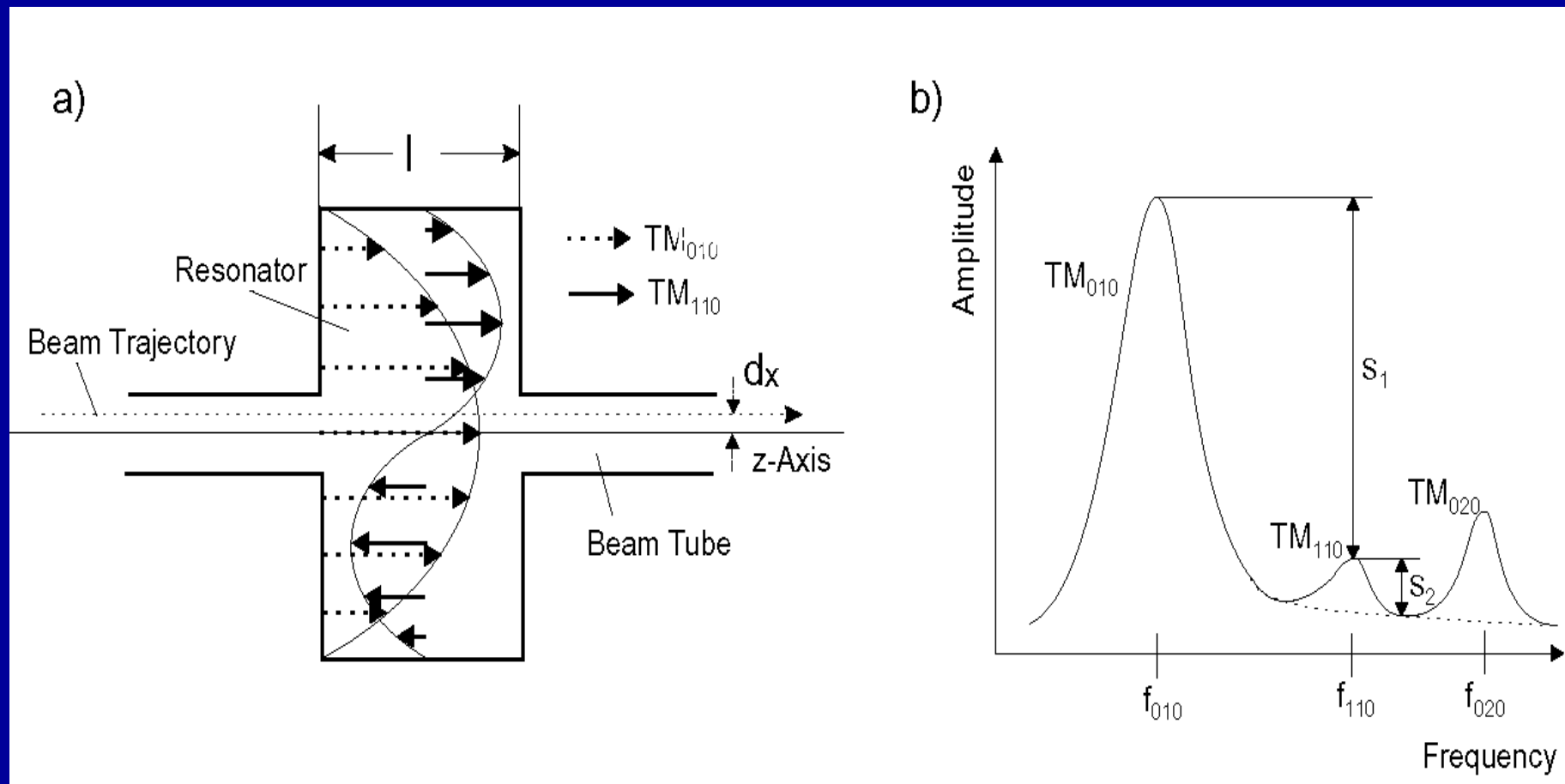
M15 “can”



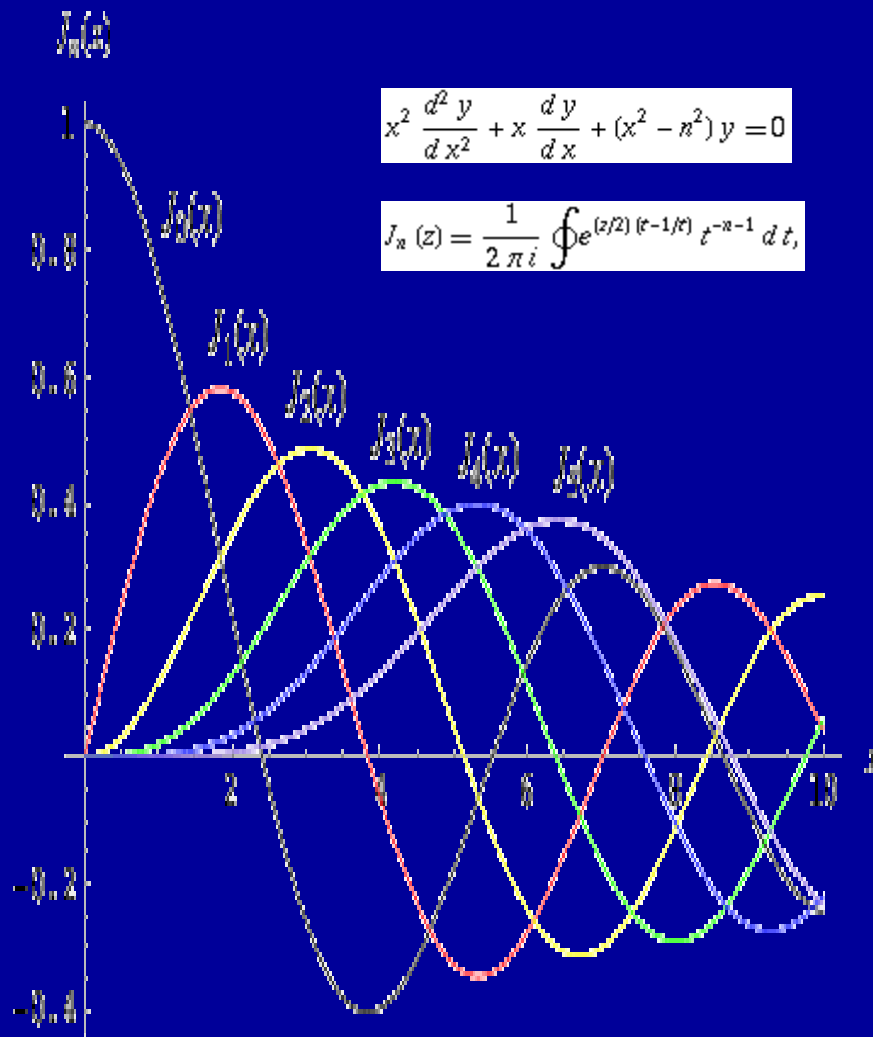
Pillbox Cavity



Cavity Response



Cavity BPM: X, Y, and I

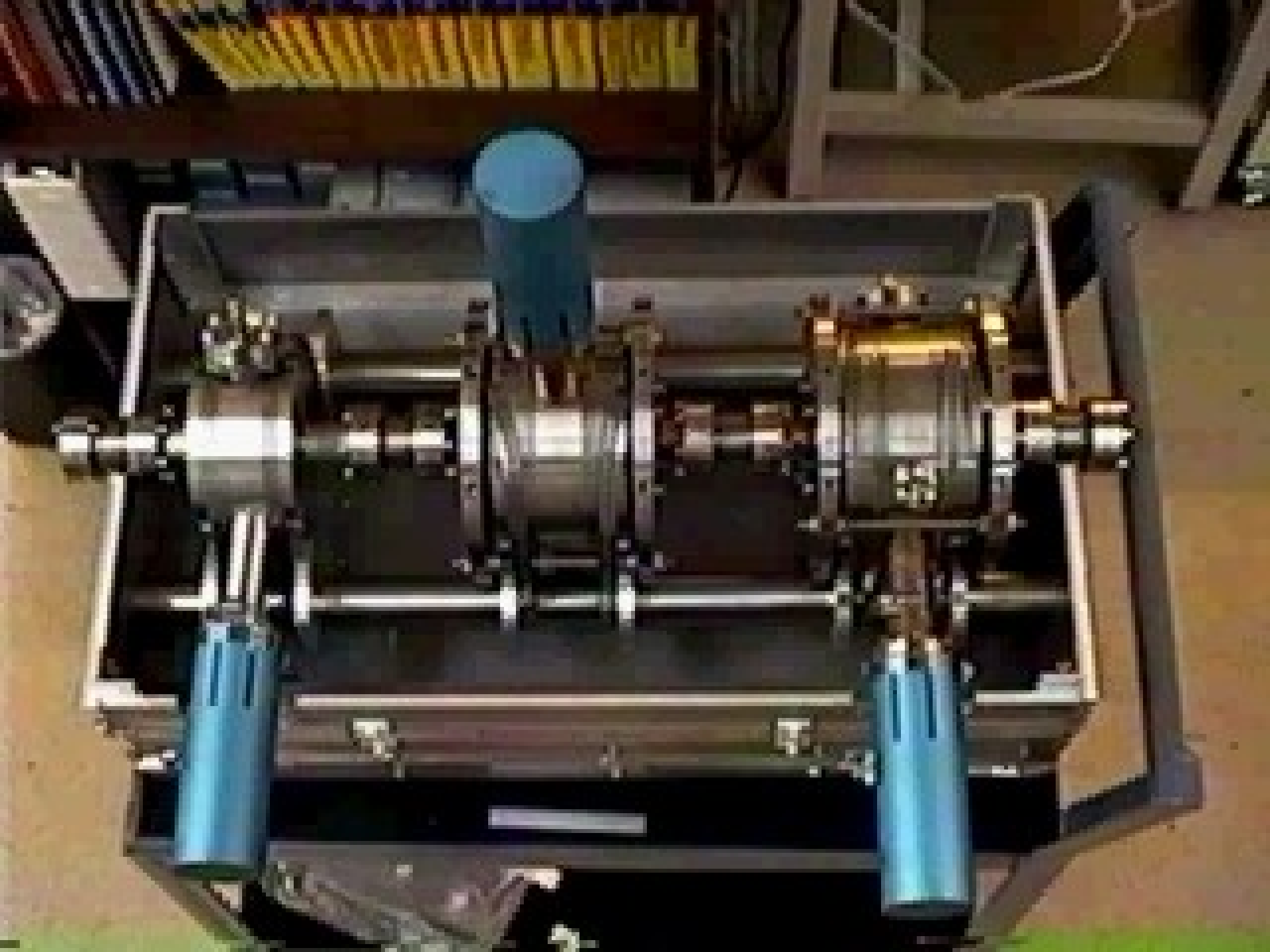


TM010 Mode for I

TM110 Mode for X & Y

Slugs provide proper excitation, reducing TM010 x-talk

Nominal output: 54nV-
uA/um (-132 dBm)...per
MAFIA simulations



A Tutorial: Signals and Noise

- Resolution is determined by signal-to-noise ratio
- Fancy algorithms help w/ filtering/integration, etc.
but physics prevails.
- So, what is noise? Signal?
- Also, what is meant by “Dynamic Range?”
 - AKA “Linearity”

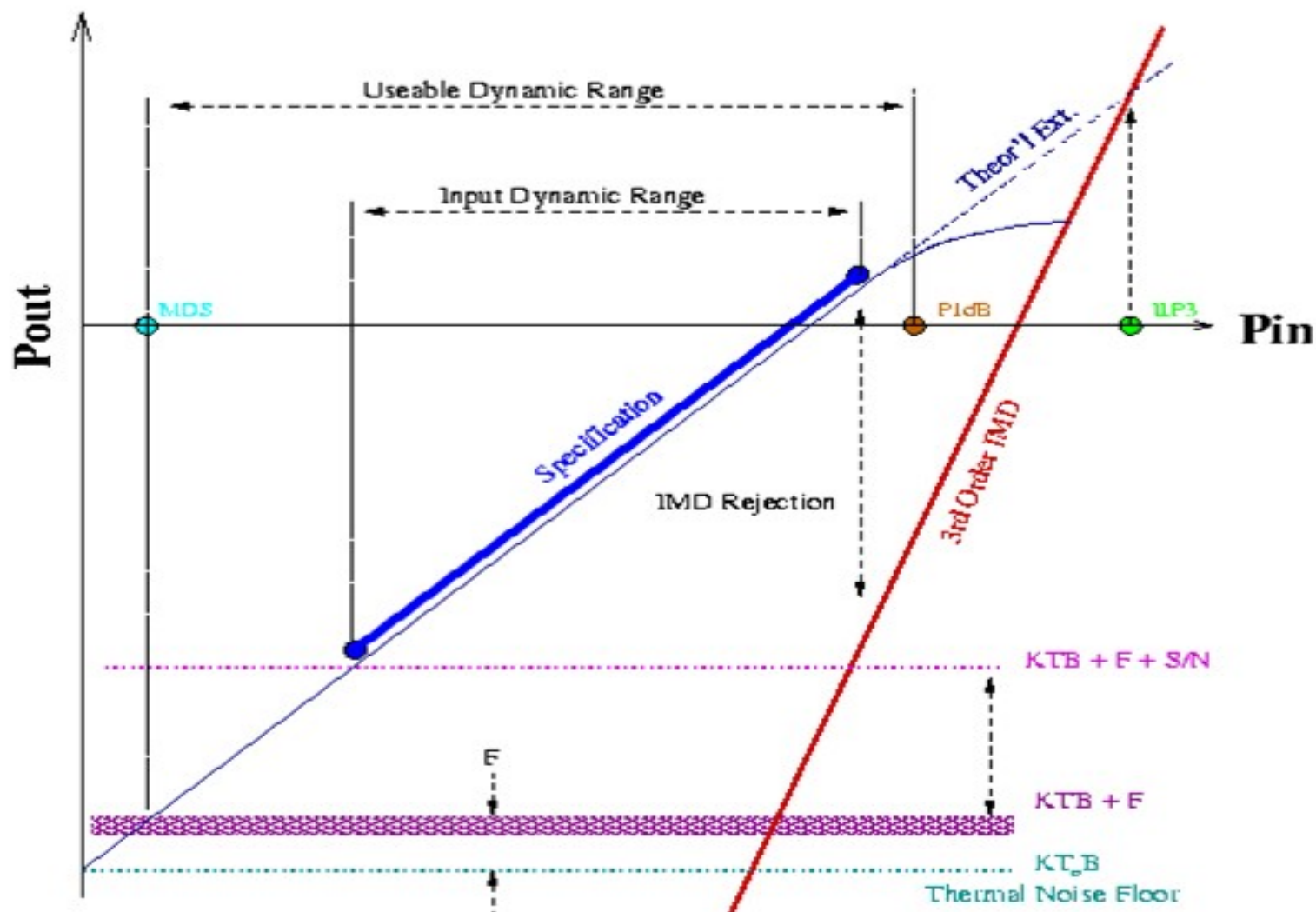
Required Input Parameters

- Dynamic Range
 - Imin and Imax
- Resolution/Accuracy
 - Determines SNR
 - Based on physics and algorithm
- Output Rate
 - Ultimate system BW and output sample rate
- Isolation
 - Crosstalk and rejection of “systematics”

Tests for these parameters are relatively standardized and routine!!

Receiver Dynamic Range Parameters

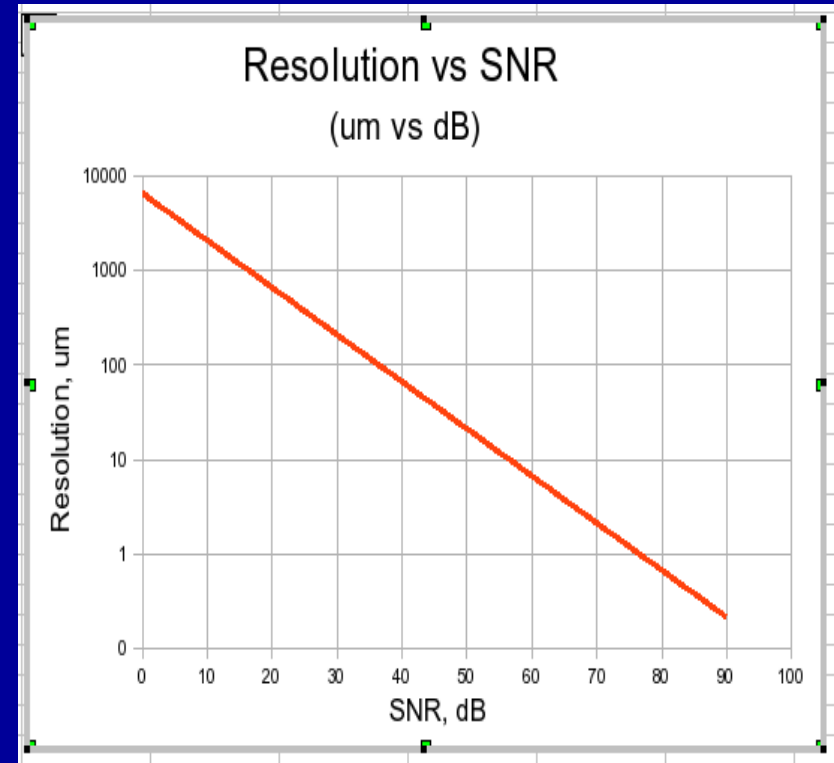
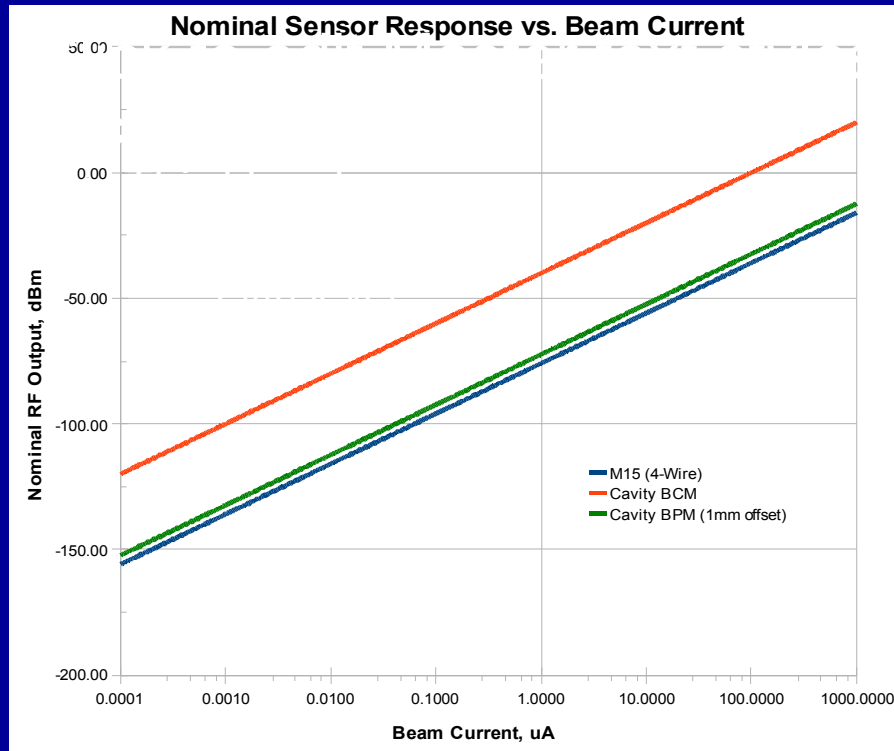
(All quantities in dB)



Rx Design Parameters (Signal Level)

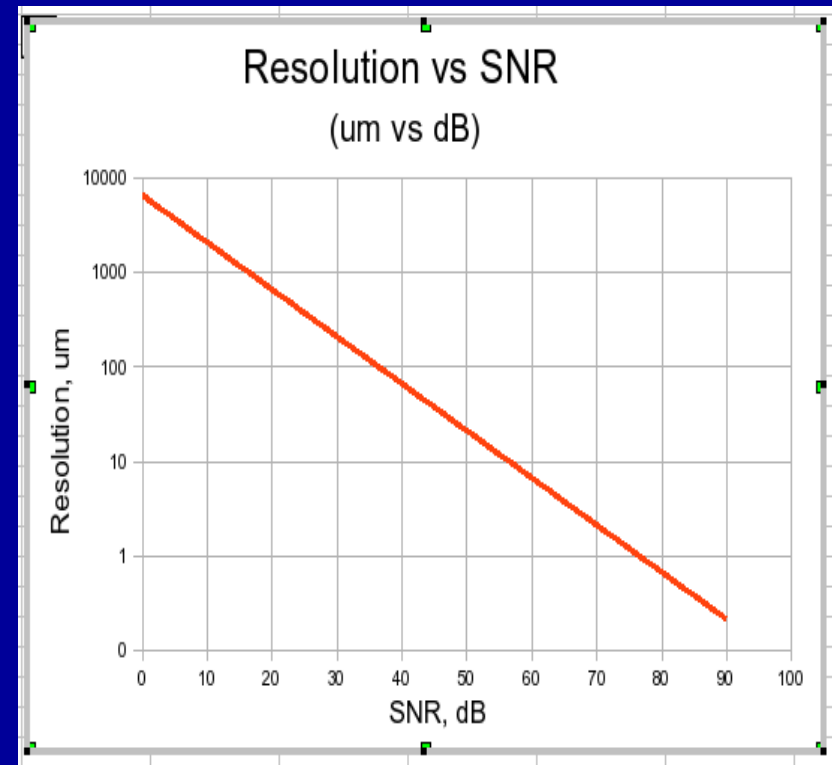
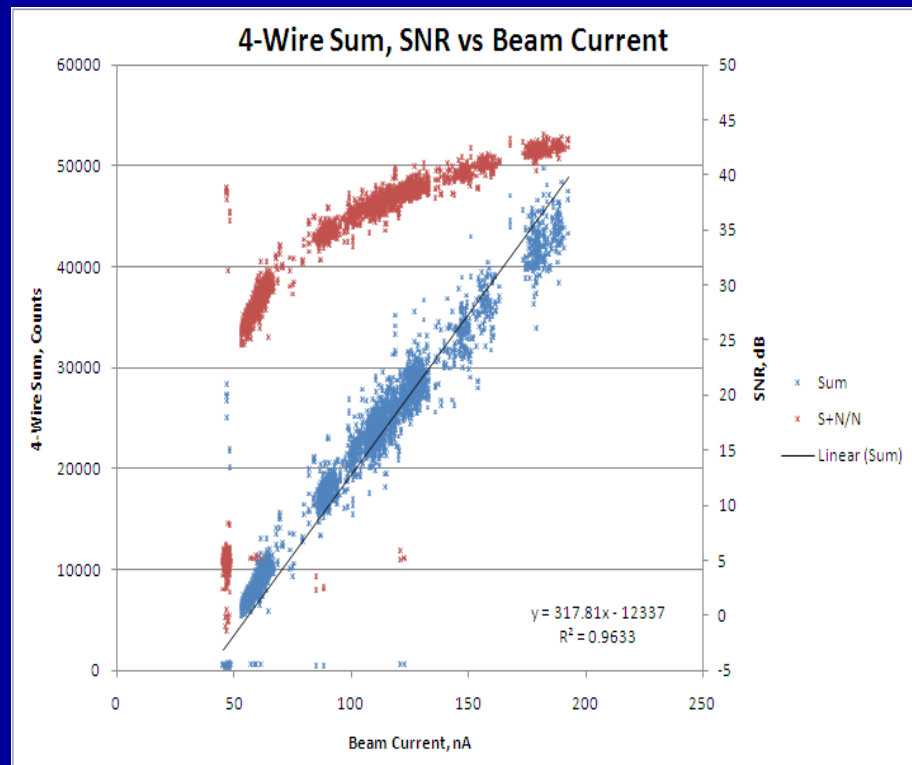
**Determine Sensor output
for given beam conditions**

**Match Rx with sensor and
SNR requirements (mostly
driven by THD specs.).**

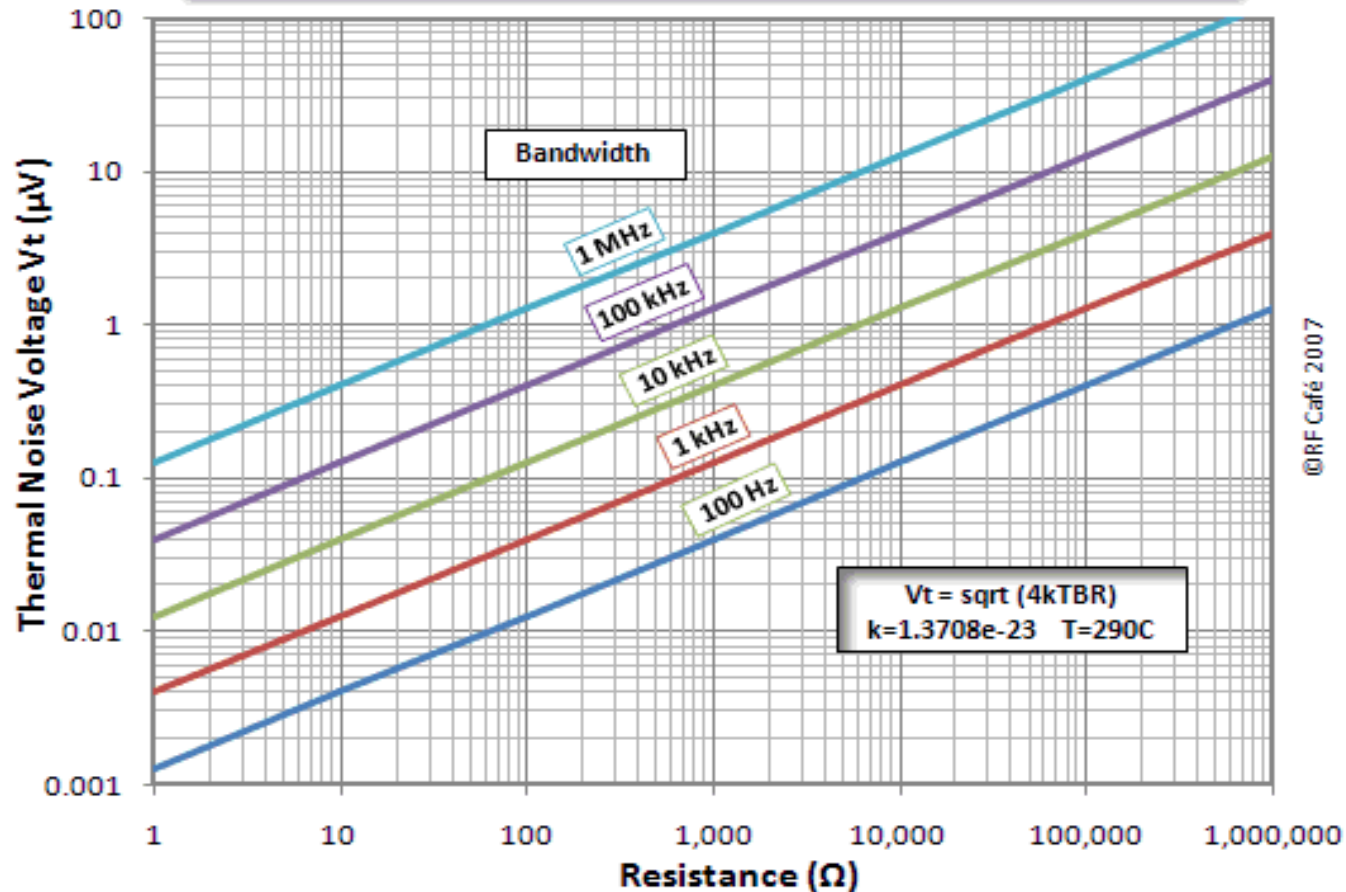


Example: Recent Beam Test

M15 w/ Prototype Electronics, 5/11



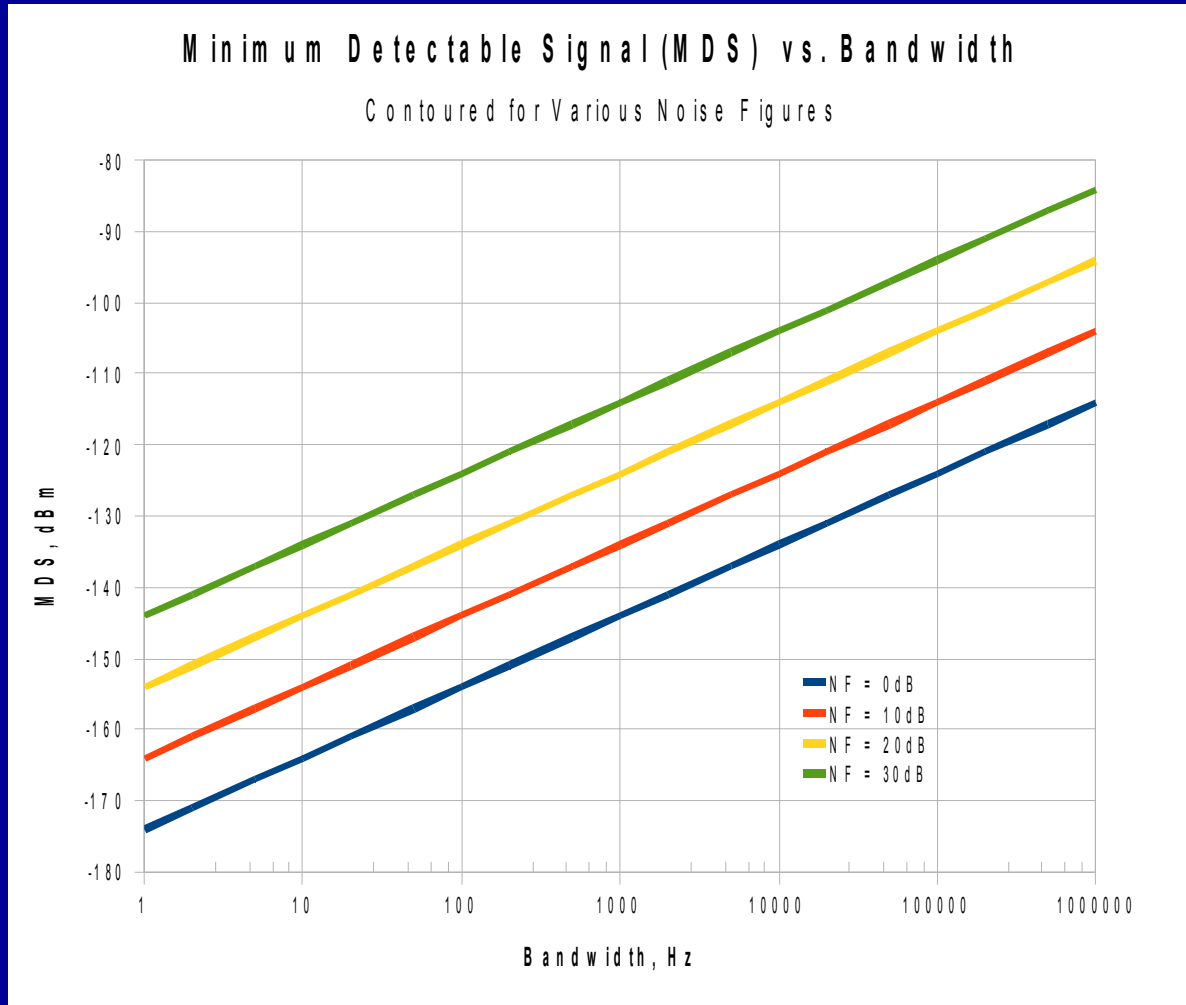
Thermal Noise as a Function of Resistance and Bandwidth



$$\begin{aligned}
 kTB &= (1.38 \times 10^{-23} \text{ J/K})(293\text{K})(1 \text{ Hz}) \\
 &= 4.057 \times 10^{-21} \text{ W} = -174 \text{ dBm} \\
 &\text{or } -174 \text{ dBm/Hz in terms of PSD.}
 \end{aligned}$$

$$F = \frac{\left(\frac{S}{N}\right)_{in}}{\left(\frac{S}{N}\right)_{out}} = \frac{S_{in}}{N_{in}} \times \frac{N_{out}}{S_{out}}$$

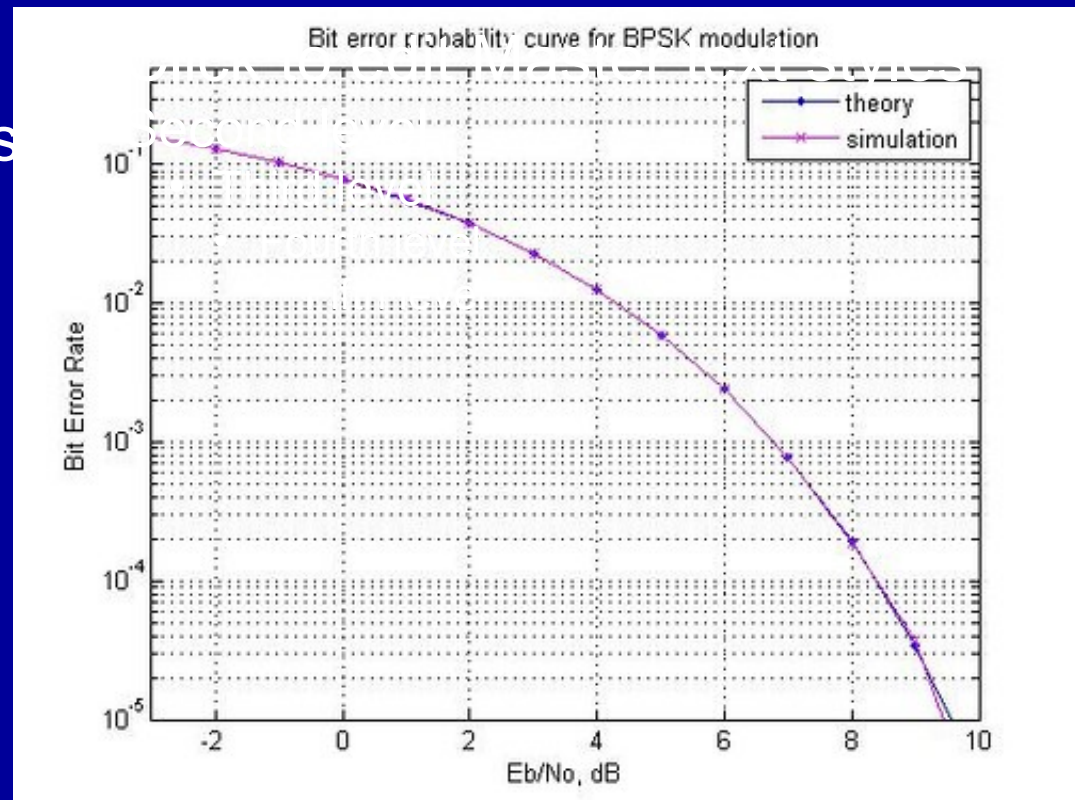
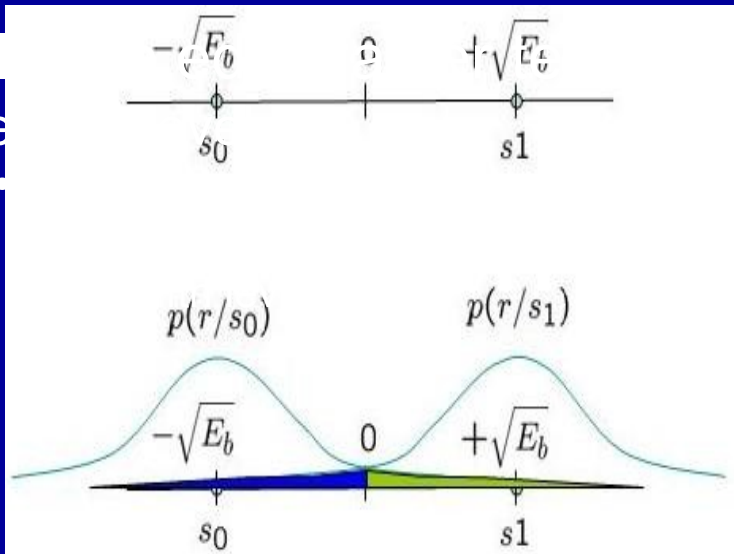
Noise Floor for 50 Ohm System



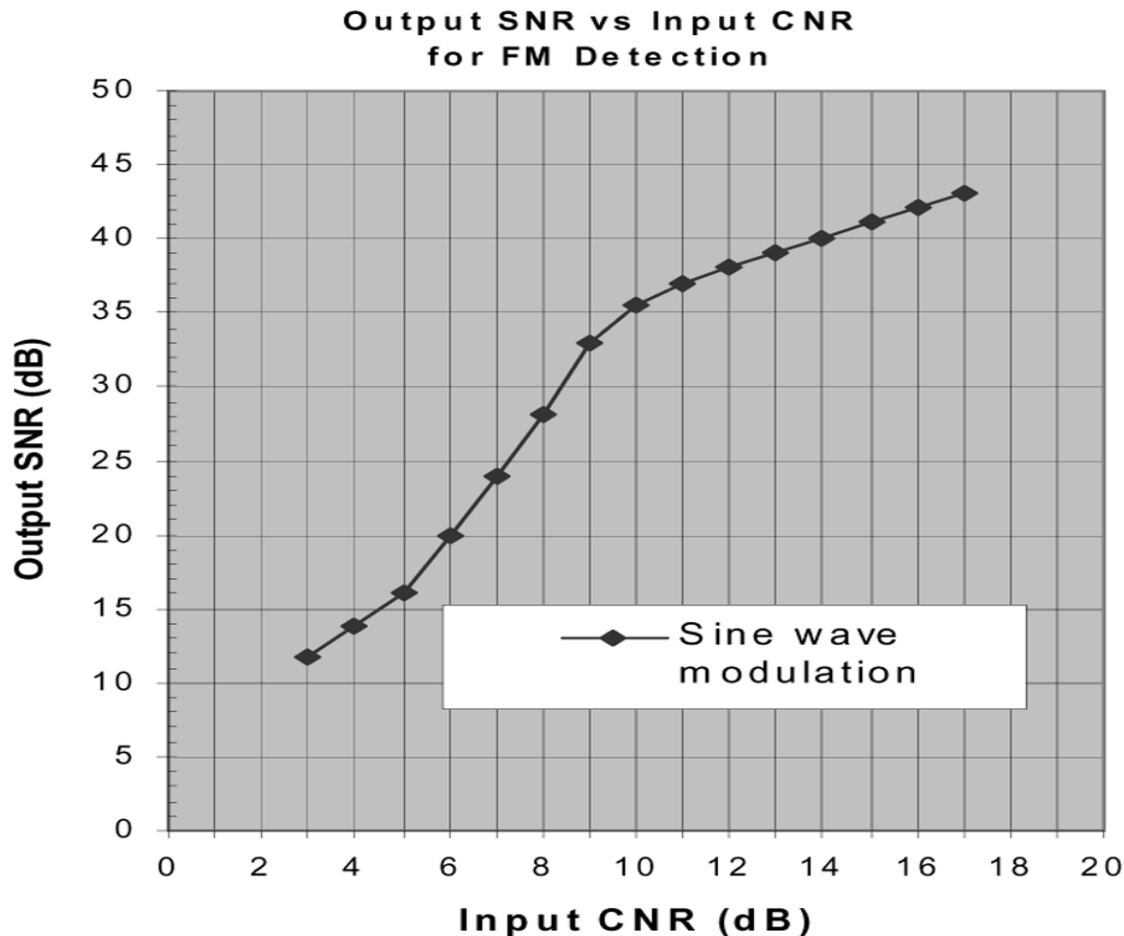
Phase Measurement Drives SNR

SNR determines ability to resolve +/- phase at low end:

Identical to BPSK BER analysis



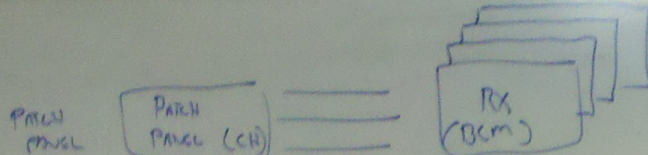
Extra Baggage....



Extra SNR (~ 13 dB)
is needed for angular
demodulation, since
it is non-linear
("threshold ing")

$BW = 100 \text{ kHz}$
 $I_{min} = 150 \text{ mA}$ (70 mA)
 $I_{max} = 200 \text{ mA}$ (170 mA)
 $SUR_{min} = ?$
 $I_{solutions} = ?$
 $\# \text{ output DITS} = ?$

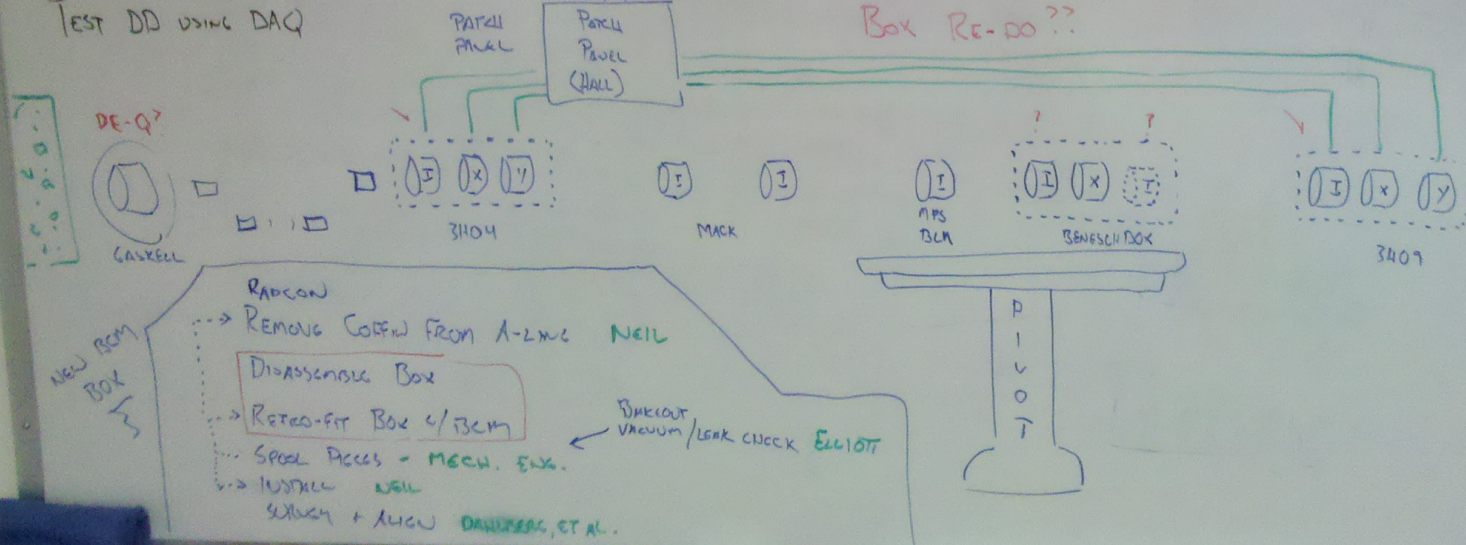
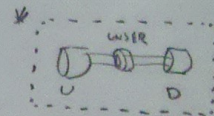
Train Hall C Personnel $\Rightarrow$ D.D. Test?
 PSD of D. Mack e^-
 Output Calc Rate (DD) = 1 k/sps
 ADC $\sim 1/4 \sim 1 \text{ MHz}$?
 Test DD using DAQ



PROTOTYPE
 MS6 3H04 - 6000(?)
 '0602 3H09 - 0800 $\rightarrow$ FIX
 MS6 *3 - 6000

	HARDWARE	e^-	
62P	2 CANS, MINI MITS (BCMS EXIST)*	2 RECEIVERS 1 RECEIVER 1 RECEIVER	RED P-TYPE
IIC	QWEAK	5x2CH RX (10CH)	
GASKELL		1 HARPER RX (EXISTS)	
MPS/BCA	1 BCM	e^- EXISTS	

* CAN WE USE EXISTING MPS BCM
 (FROM PRIOR GINDAL) IN PLACE OF
 BOX RE-DO??

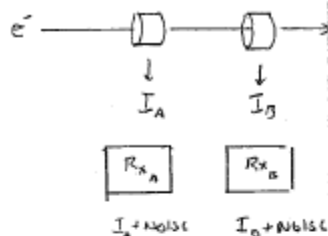


RADCON
 $\rightarrow$ REMOVE COFFIN FROM A-ZMC NEIL
 DISASSEMBLE BOX
 $\rightarrow$ RETRO-FIT BOX w/ BCM
 SPOOL ACCESS - MECH. ENG.
 $\rightarrow$ INSURE NEIL
 SULLIVAN + ALAN DALLMEIER, ET AL.

QWEAK Example

BCM Error Analysis

$$DD = \frac{I_A - I_B}{I_A + I_B}$$



Assume:

Each entity is measured with independent receiver,
so errors are random and uncorrelated.
Expressed as functional errors:

$E[X] = \langle X \rangle =$ "Expected Value of X"

$$I_A = \langle I_A \rangle \pm \frac{\partial I_A}{I_A}$$

$$I_B = \langle I_B \rangle \pm \frac{\partial I_B}{I_B}$$

Assume "Similar" Receivers, so

$$\frac{\partial I_A}{I_A} \sim \frac{\partial I_B}{I_B} = \delta$$

And,

$$DD = \frac{I_A - I_B}{I_A + I_B} = \frac{\langle I_A \rangle (1 \pm \delta) - \langle I_B \rangle (1 \pm \delta)}{\langle I_A \rangle (1 \pm \delta) + \langle I_B \rangle (1 \pm \delta)}$$

$$DD_{small} = \frac{\langle I_A \rangle - \langle I_B \rangle}{\langle I_A \rangle + \langle I_B \rangle} [1 - 2|\delta|]$$

So,

$$DD = \frac{\langle I_A \rangle - \langle I_B \rangle}{\langle I_A \rangle + \langle I_B \rangle} [1 \pm 2|\delta|]$$

$$\frac{\partial DD}{DD} = 2|\delta|$$

For 1%,

$$2|\delta| = 0.01$$

$$|\delta| = 0.005 = \frac{\partial I_A}{I_A} = \frac{\partial I_B}{I_B}$$

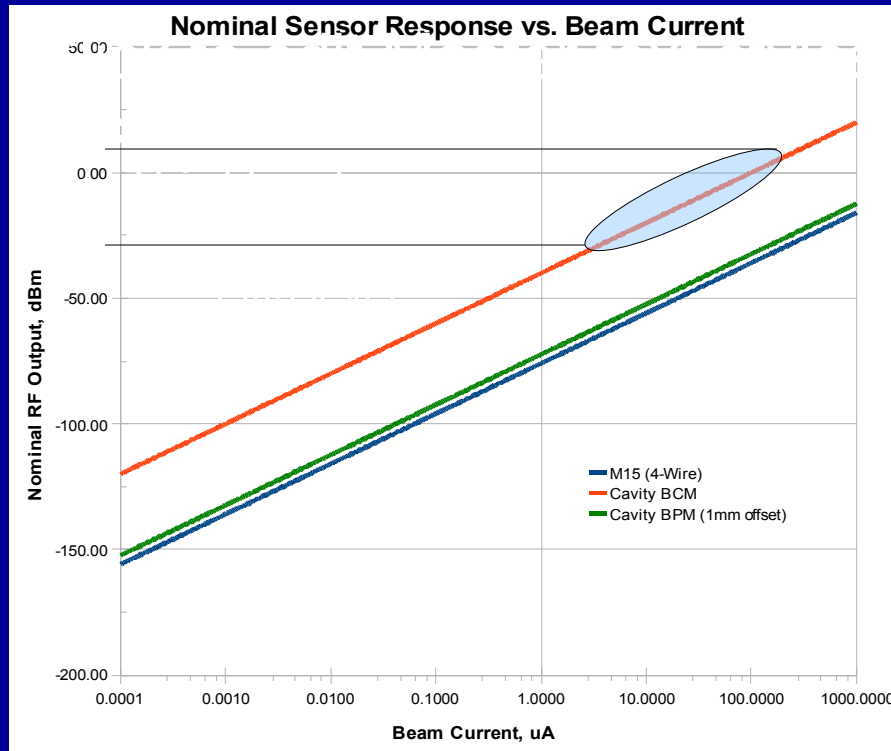
$$20 \log [0.005] = -46 \text{ dB}$$

Hence, 1% DD $\Rightarrow$ 46 dB SNR.

QWEAK Design Parameters

**Determine Sensor output
for given beam conditions**

**Match Rx with sensor and
SNR requirements for
double-difference of 1%**



Noise BW = 100 kHz

KTB = -124 dBm (perfect 290K Rx)

Min SNR = -20 dBm - (-124) = 104dB

	2	2	2	0	2	2
	4	4	4	0	4	4
	6	6	6	0	6	6
	8	8	8	0	8	8
	9	9	9	0	9	9
	10	10	10	0	10	10
	11	11	11	0	11	11
	12	12	12	0	12	12
	13	13	13	0	13	13
	14	14	14	0	14	14
	15	15	15	0	15	15
	16	16	16	0	16	16
	17	17	17	0	17	17
	18	18	18	0	18	18
	19	19	19	0	19	19
	20	20	20	0	20	20
	21	21	21	0	21	21
	22	22	22	0	22	22
	23	23	23	0	23	23
	24	24	24	0	24	24
	25	25	25	0	25	25
	26	26	26	0	26	26
	27	27	27	0	27	27
	28	28	28	0	28	28
	29	29	29	0	29	29
	30	30	30	0	30	30
	31	31	31	0	31	31
	32	32	32	0	32	32
	33	33	33	0	33	33
	34	34	34	0	34	34
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	36	36	36	0	36	36
	37	37	37	0	37	37
	38	38	38	0	38	38
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	40	40	40	0	40	40
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	42	42	42	0	42	42
	43	43	43	0	43	43
	44	44	44	0	44	44
	45	45	45	0	45	45
	46	46	46	0	46	46
	47	47	47	0	47	47
	48	48	48	0	48	48
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	51	51	51	0	51	51
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	60	60	60	0	60	60
	61	61	61	0	61	61
	62	62	62	0	62	62
	63	63	63	0	63	63
	64	64	64	0	64	64
	65	65	65	0	65	65
	66	66	66	0	66	66
	67	67	67	0	67	67
	68	68	68	0	68	68
	69	69	69	0	69	69
	70	70	70	0	70	70
	71	71	71	0	71	71
	72	72	72	0	72	72
	73	73	73	0	73	73
	74	74	74	0	74	74
	75	75	75	0	75	75
	76	76	76	0	76	76
	77	77	77	0	77	77
	78	78	78	0	78	78
	79	79	79	0	79	79
	80	80	80	0	80	80
	81	81	81	0	81	81
	82	82	82	0	82	82
	83	83	83	0	83	83
	84	84	84	0	84	84
	85	85	85	0	85	85
	86	86	86	0	86	86
	87					

-55.39	dBm
-6.00	dBm
49.39	dB

Input Field

Noise Figure

Gain: Passband

IIP3

PIdB

Return Loss

Pin Passband

Input Noise BW

Input Noise Temperature

Input Noise Level

Required C/N

Required Sensitivity



-6.00

50.00

290.00

-124.0

13.00

-60.00

Heliax

"@1437

Amp

Atten

Mixer

Atten/LPF

VGA

ADC

"@1 MHz

32.00

3.50

1.00

6.00

11.00

7.00

24.00

dB

-32.00

18.00

-1.00

-6.00

-11.00

40.00

0.00

dB

200.00

13.10

200.00

15.00

200.00

20.00

200.00

dBm

200.00

9.00

200.00

5.00

200.00

3.00

200.00

dBm

3.00

24.00

2.00

12.00

22.00

20.00

0.00

dB

dBm

dB-Hz

(System IF BW)

K

(IEEE definition = 290K)

dBm

dB

(Modulator / BER -dependent...see BER sheet)

dBm

(From "Specifications" or "Standards")

Calculation Field

System Noise Figure

System Noise Temp

System Gain: Passband

IIP3: Passband

Input Spurious-Free Dynamic Range

Pout: Passband

Output Noise Power

C/N Ratio

Saturation?

IIM3

C/I Ratio

Distortion

Total Return Loss

32.00

35.50

35.51

35.62

37.08

40.59

40.61

dB

56.62

60.12

60.13

60.24

61.71

65.22

65.23

dBK

-32.00

-14.00

-15.00

-21.00

-32.00

8.00

8.00

dB

200.00

45.10

45.10

29.87

29.87

29.84

29.84

dBm

194.67

89.07

89.06

78.83

77.86

75.50

75.49

dB

-38.00

-20.00

-21.00

-27.00

-38.00

2.00

2.00

dBm

-124.00

-102.50

-103.49

-109.38

-118.92

-75.41

-75.39

dBm

86.00

82.50

82.49

82.38

80.92

77.41

77.39

dB

NO

NO

NO

NO

NO

NO

NO

-418.00

-108.20

-108.20

-77.74

-77.74

-77.68

-77.68

dBm

412.00

102.20

102.20

71.74

71.74

71.68

71.68

dB

0.01

0.01

0.01

0.03

0.03

0.03

0.03

%

3.00

27.00

29.00

41.00

63.00

83.00

83.00

dB

Calculated Receiver Sensitivity:

-70.39

dBm

Required Receiver Sensitivity:

-60.00

dBm

Margin:

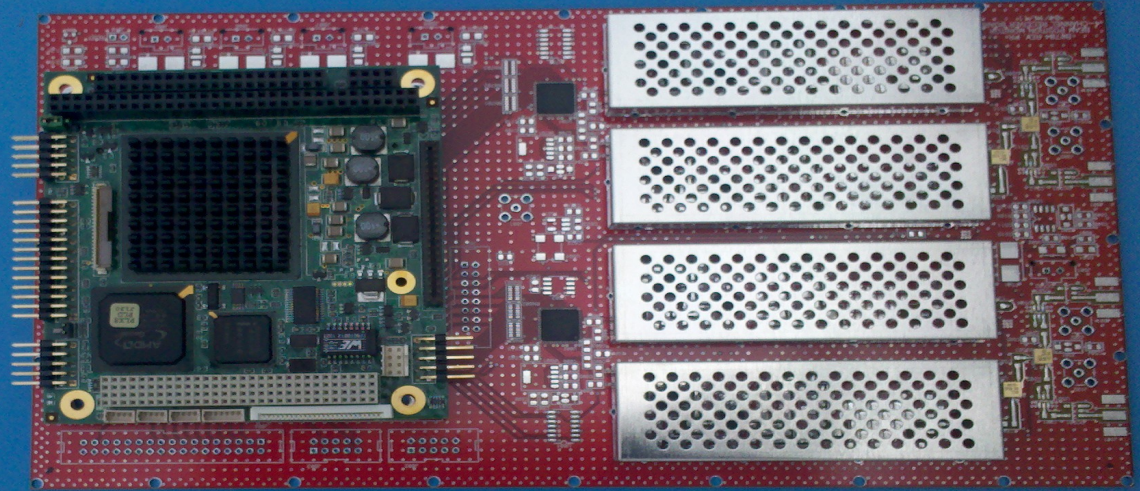
10.39

dB

Rx Preliminary Performance

Quantity (1 Hz BW)		Max Gain	Min Gain
Noise Figure		2 dB	3 dB
Channel Gain		108 dB	48 dB
Input IP3		-70 dBm	-57 dBm
MDS (ADC- limited)		-170 dBm	-133 dBm
F.S. Input		-72 dBm	-37 dBm
Output SNR	1 Hz	71 dB	96 dB
	100 kHz	21 dB	46 dB
0.1% THD Range		10 dB-Hz	30 dB-Hz
1% THD Range		30 dB-Hz	50 dB-Hz
(dB-Hz = MDS - 20log(THD) - IIP3)			
Signal Level	M15 (4-wire sum)	BCM	Cavity BPM, 1mm
-62 dBm	5 uA	80 nA	3 uA
-112 dBm	17 nA	250 pA	11 nA
-143 dBm	505 pA	8 pA	319 pA

PCB Layout

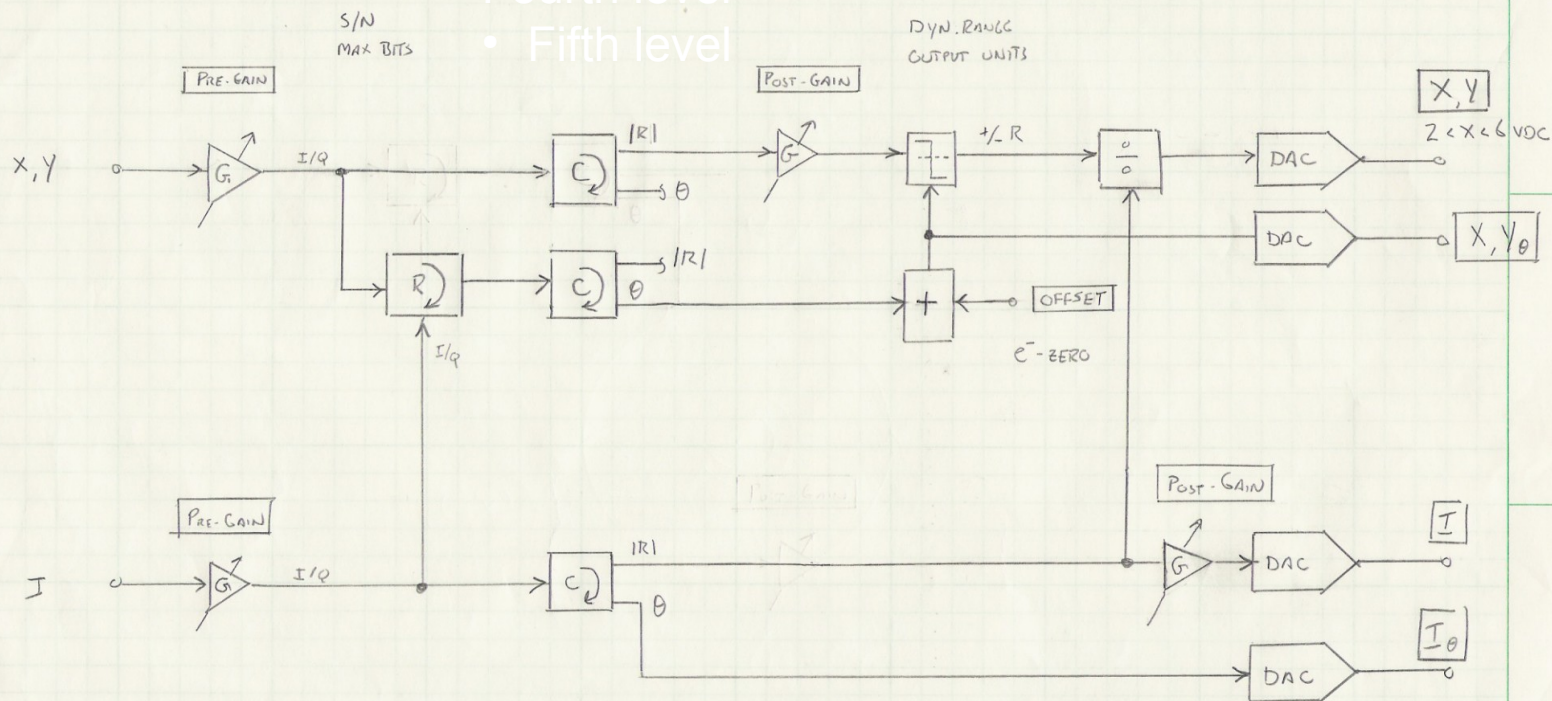


Functional Description

Click to edit Master text styles
Second level

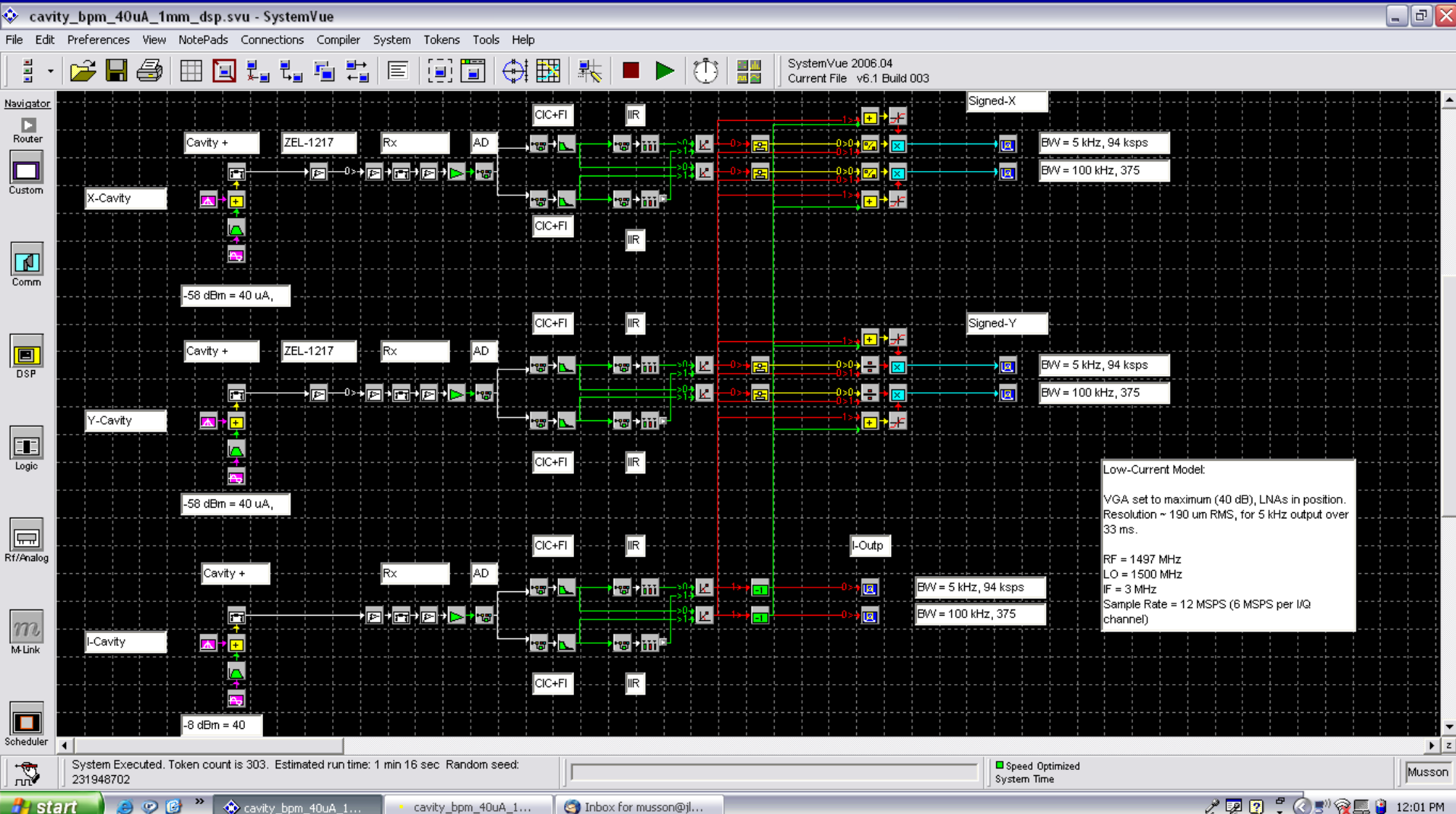
13-752 500 SHEETS, FILLER, 5 SQUARE
42-381 80 SHEETS, EYE-EASE, 5 SQUARE
42-382 100 SHEETS, EYE-EASE, 5 SQUARE
42-389 200 SHEETS, EYE-EASE, 5 SQUARE
42-392 100 RECYCLED WHITE, 5 SQUARE
42-399 200 RECYCLED WHITE, 5 SQUARE
Made in U.S.A.

- Third level
- Fourth level
- Fifth level

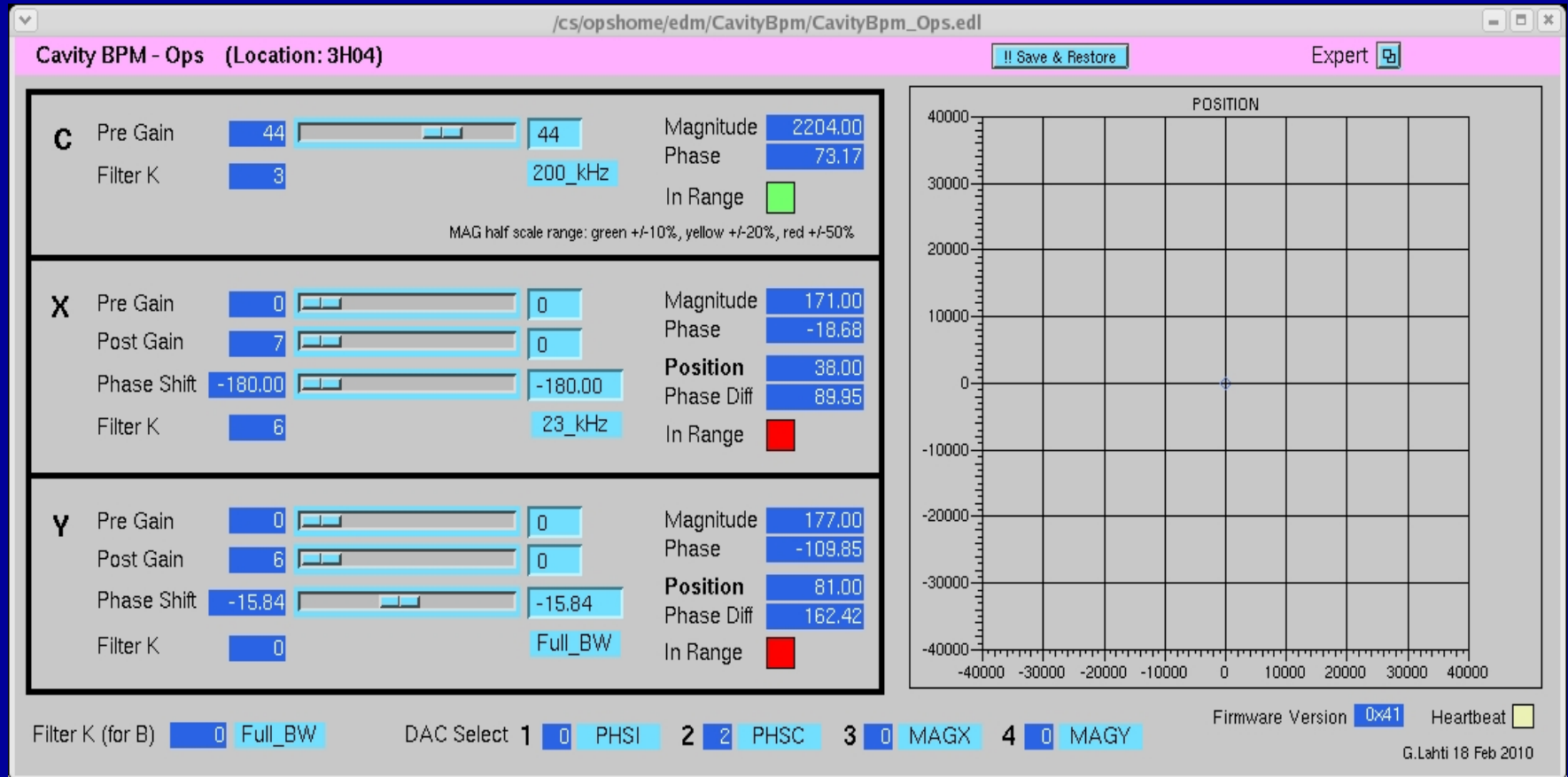


HAPPEN RECEIVER
FUNCTIONAL DIAGRAM
J. MUSSON | 4/05/05

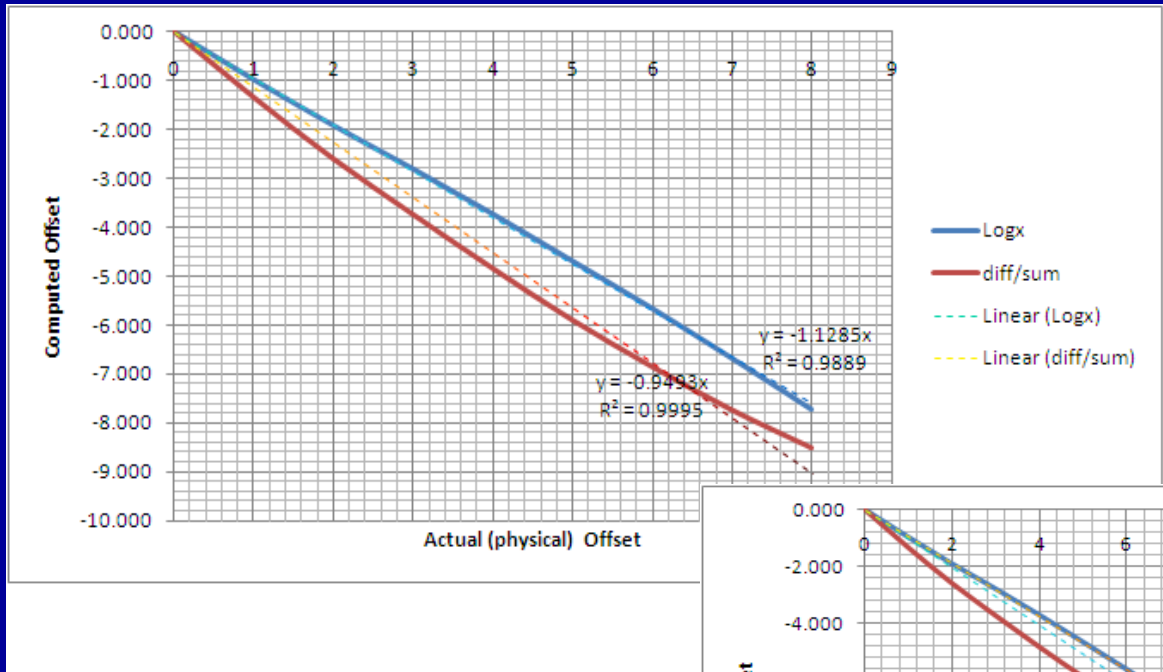
Firmware: SystemVue + AHDL



EPICS Interface / Calibration

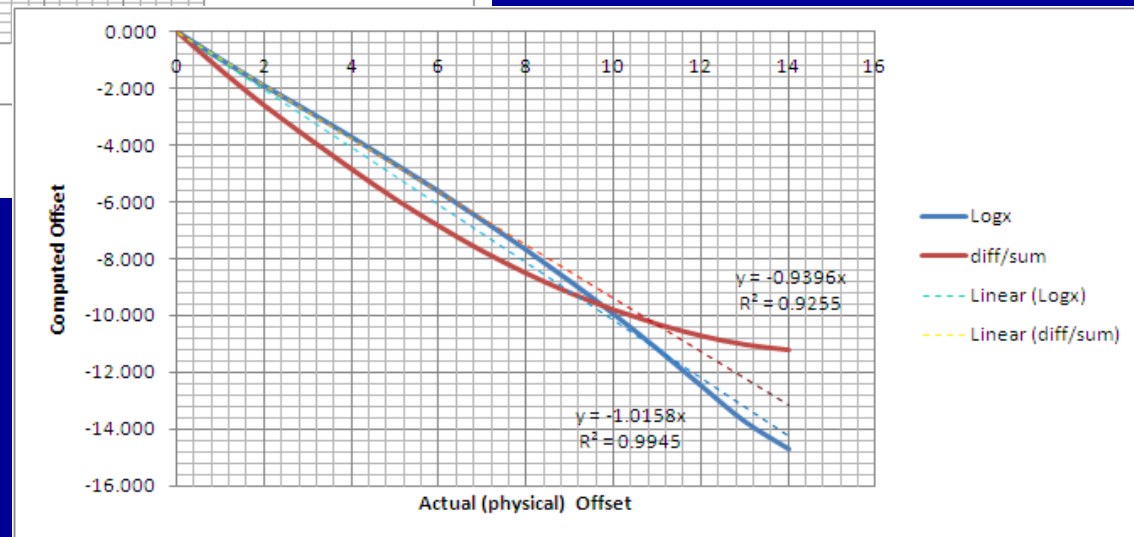


Algorithm Dependence



$$Position = \frac{\Delta}{\Sigma}$$

$$Position = \log \frac{X_a}{X_b}$$

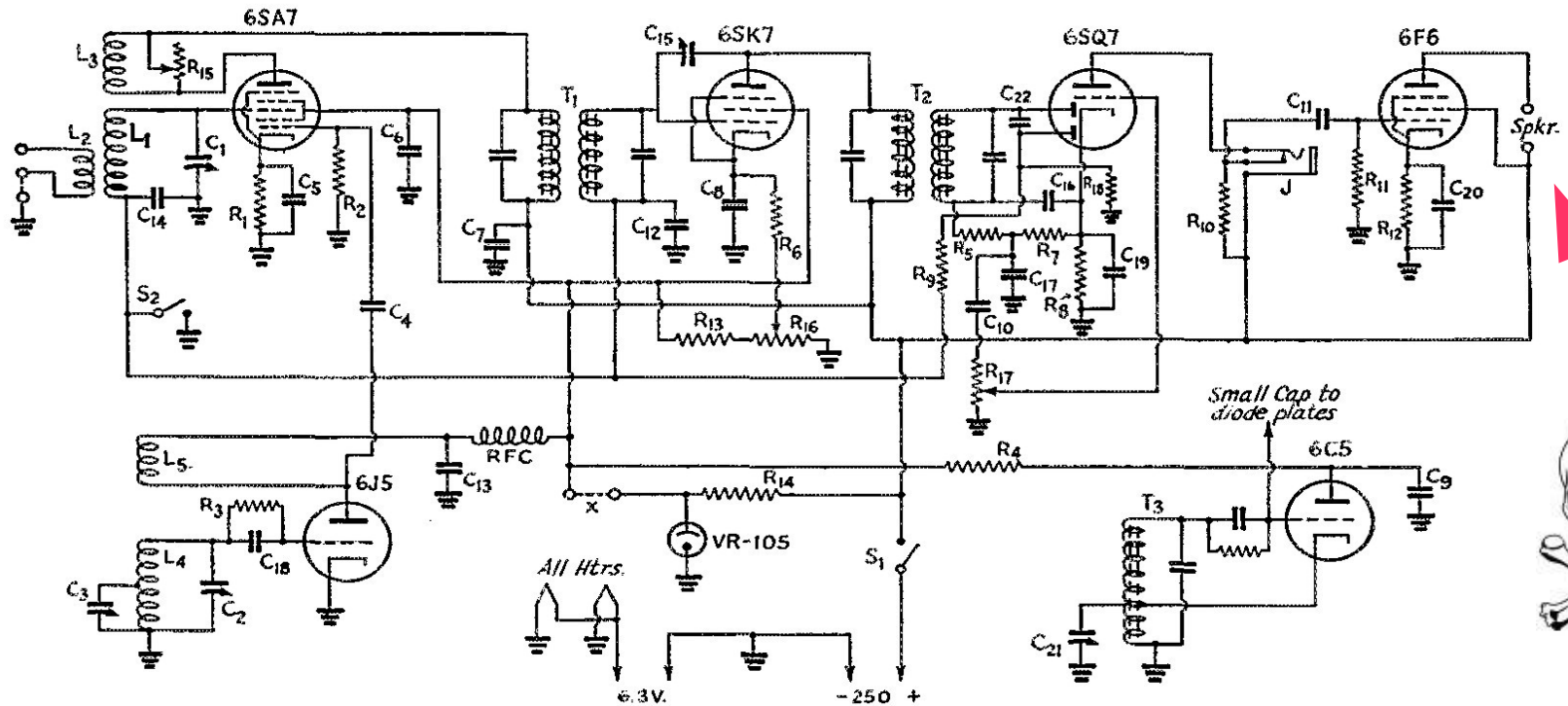


Accuracy and noise sensitivity are affected



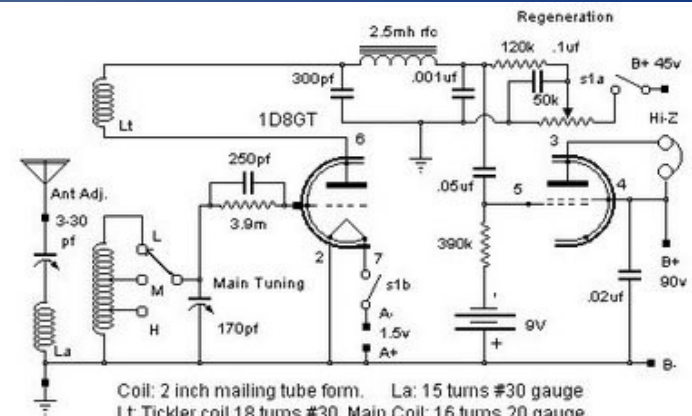
Analog Receiver Architecture





Look in your parents' attic!

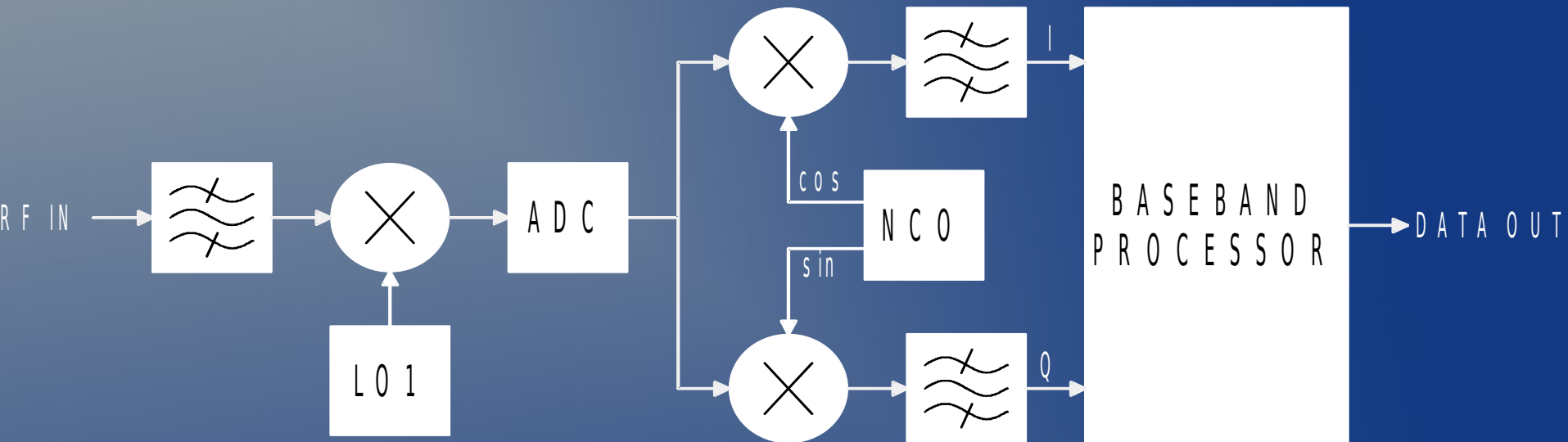
KC9KEP, ARRL



1D8 Shortwave Tube Radio (c) 2003, D. Schmacher



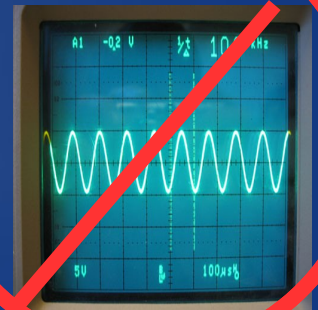
Digital Receiver Architecture



Nearly always integer math!!!

$\sqrt{x}$

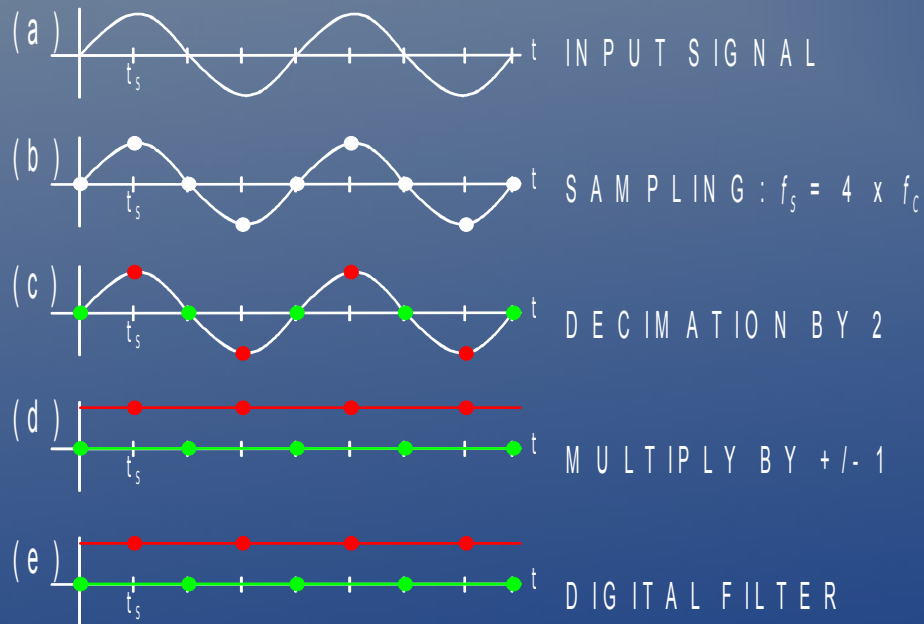
Arctan



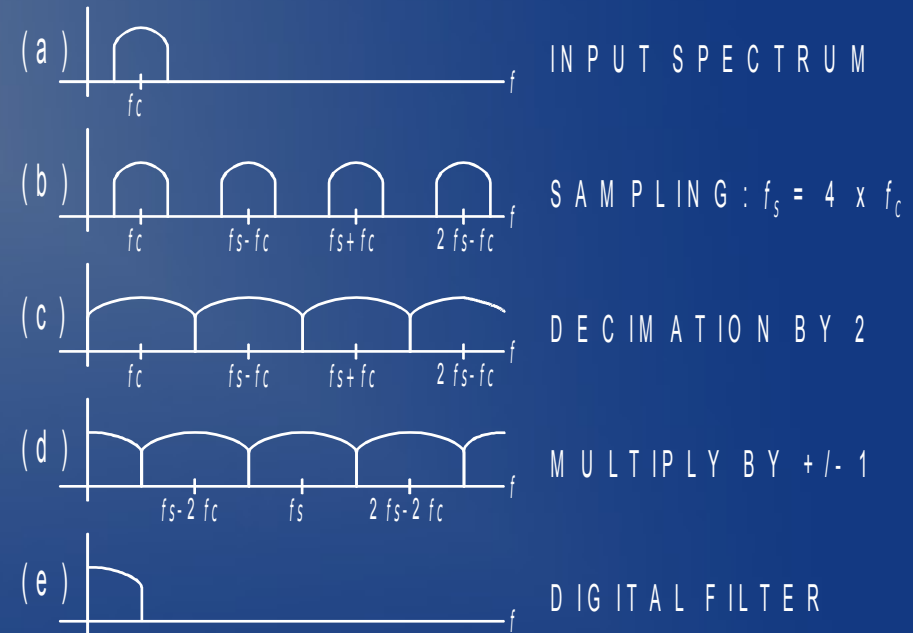


I/Q Sampling / Detection

Time Domain



Frequency Domain

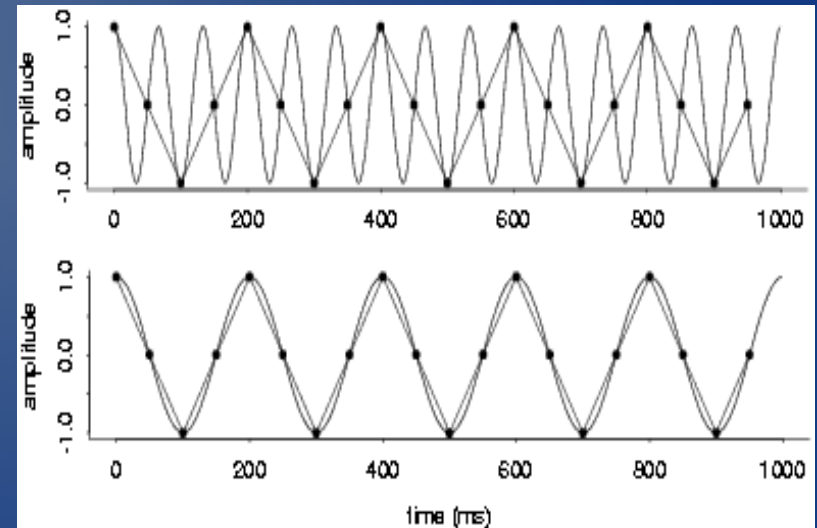
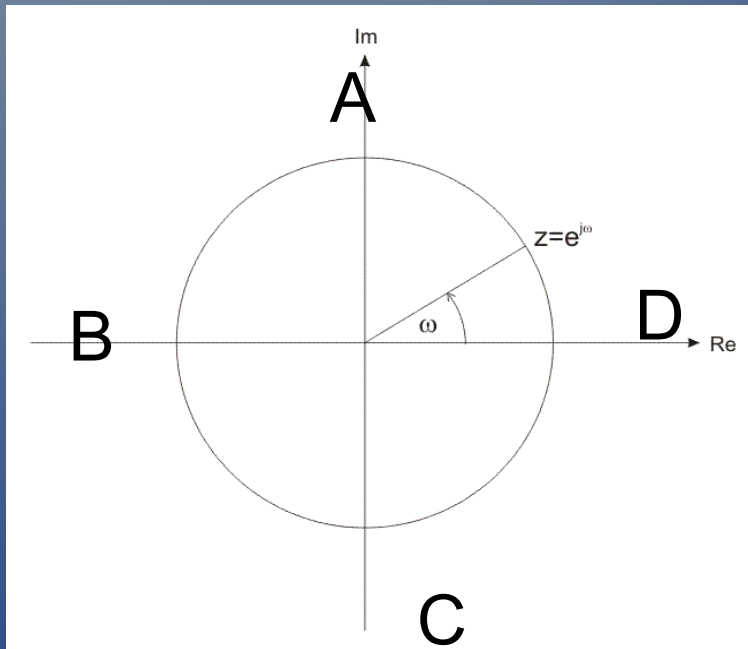


Think of N, S, E, W cardinal point sampling



Harmonic Sampling

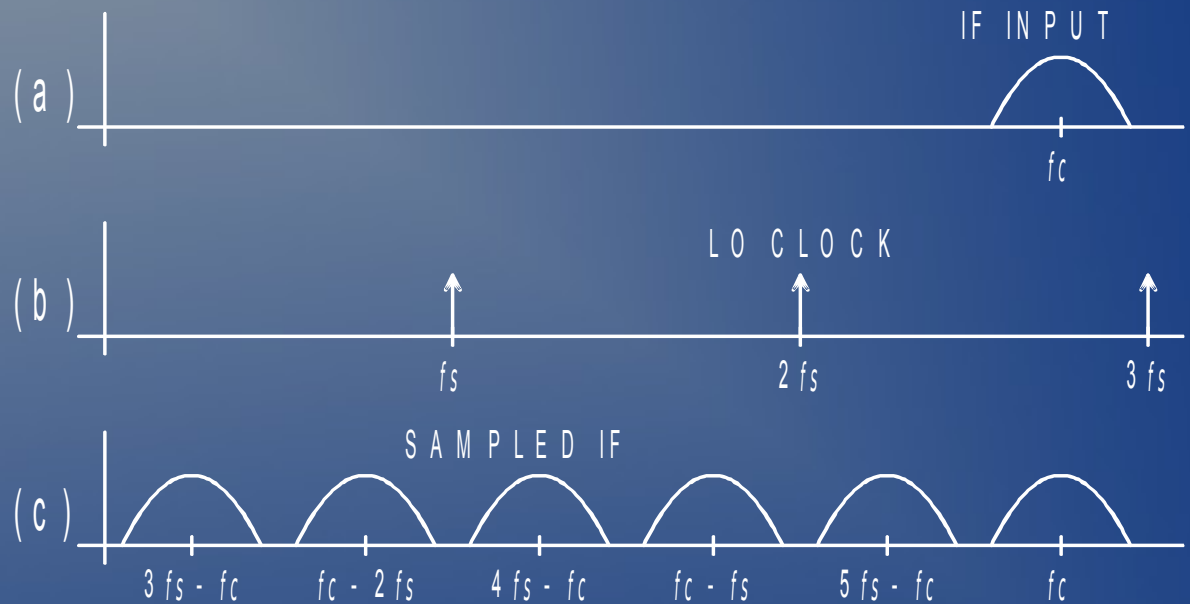
Over-rotation leads to aliasing



Scheme works for any $2\pi + (2N-1) * \pi/2$



Harmonic Sampling



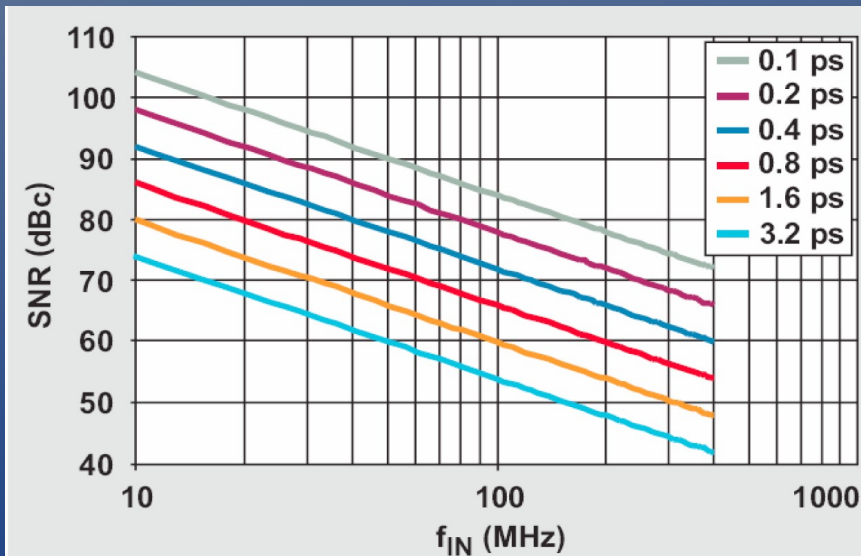
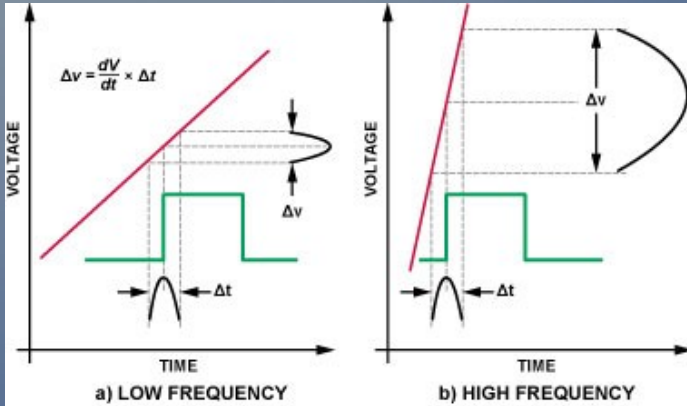
ADC Clock can be much lower than IF, to a point.....Jitter and Nyquist

Analog sampling ~ 100+ years old! Mark Kahrs, et al.

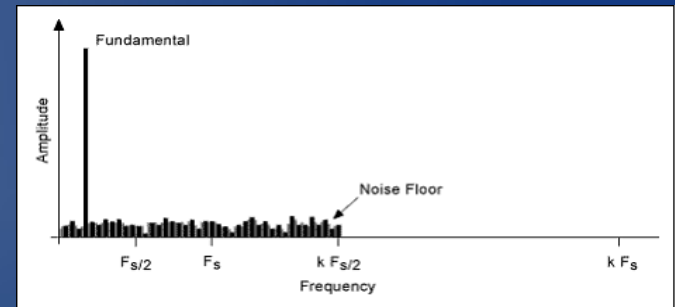
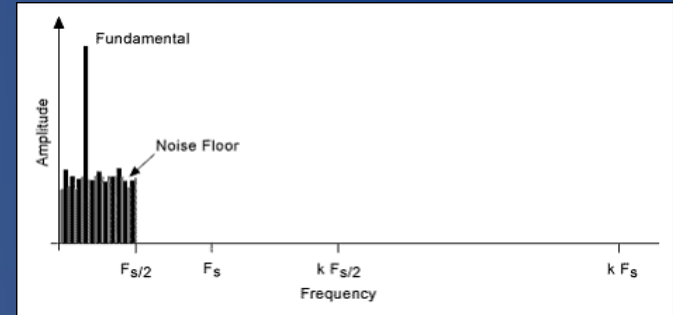


Undersampling

Cost



Benefit

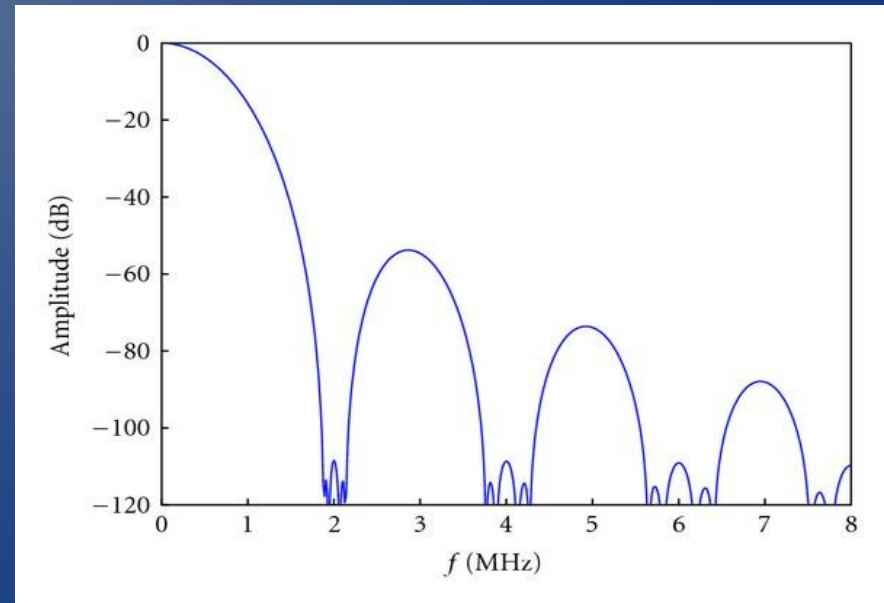
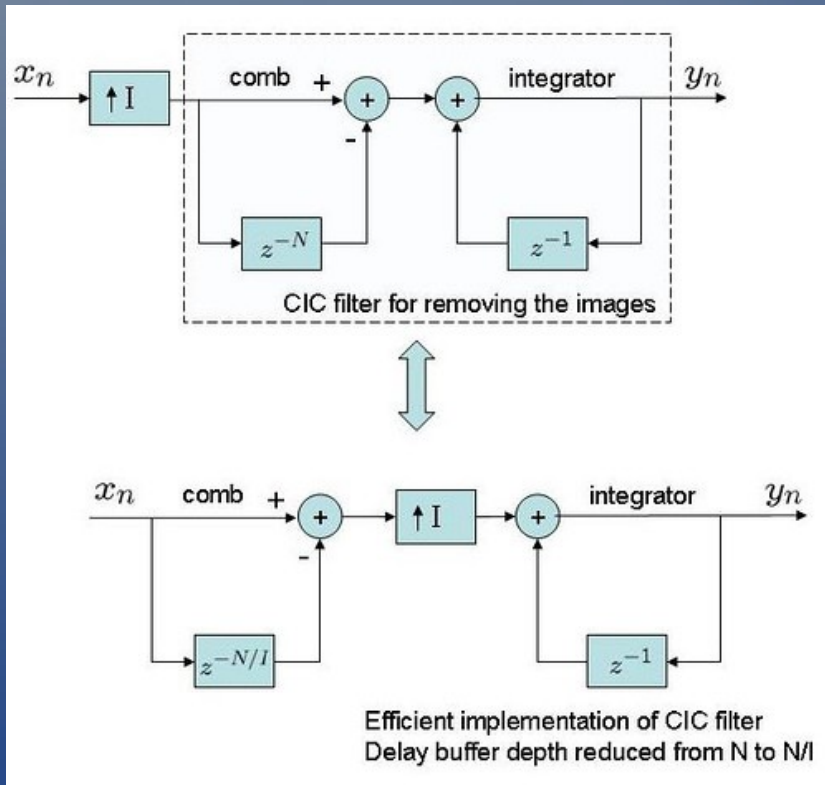
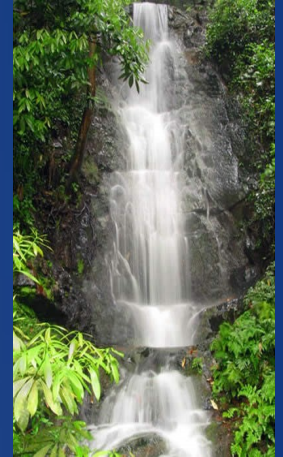


Information BW still greatly OS!



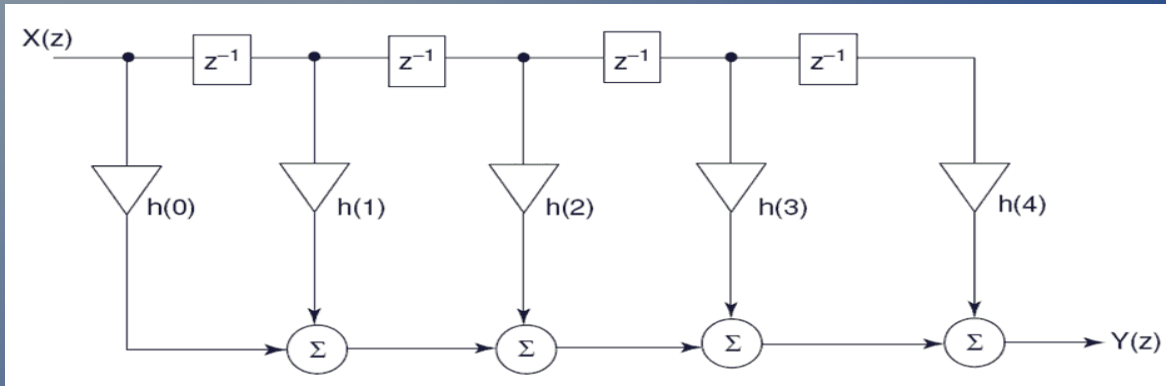
Filtering / Energy Extraction

Cascaded Integrator Comb (CIC)





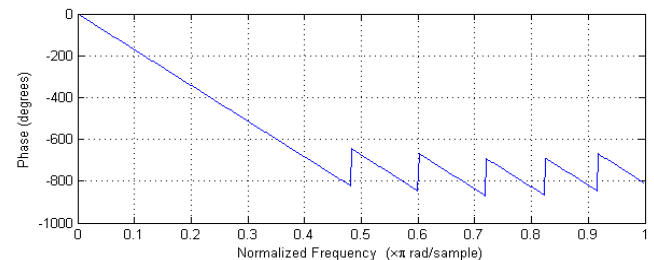
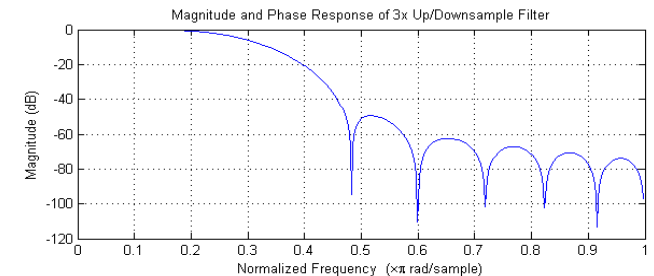
Finite Impulse Response (FIR)



Large number of taps usually required

Large latency.....Quality of Service

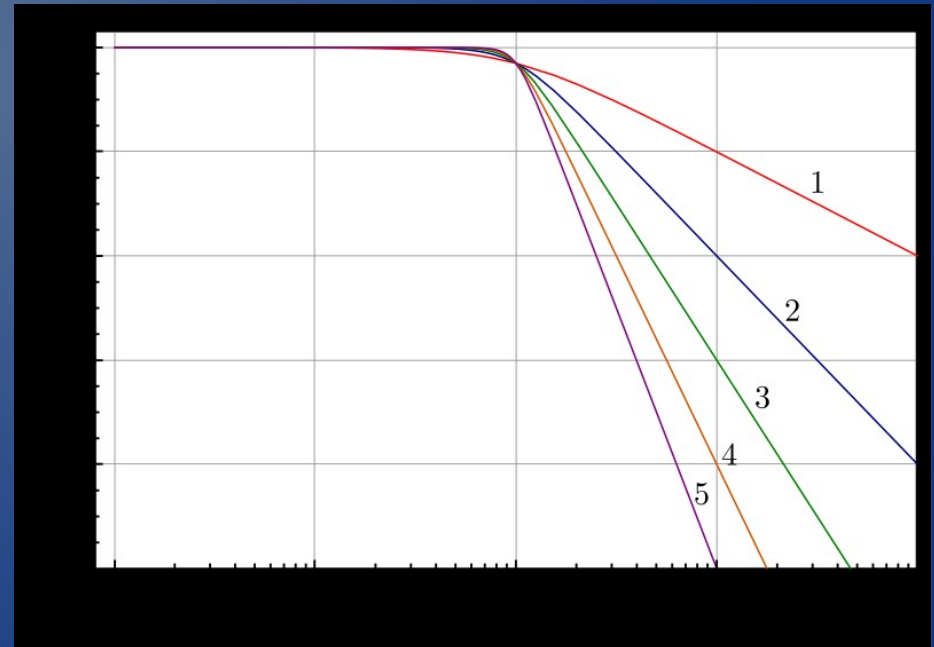
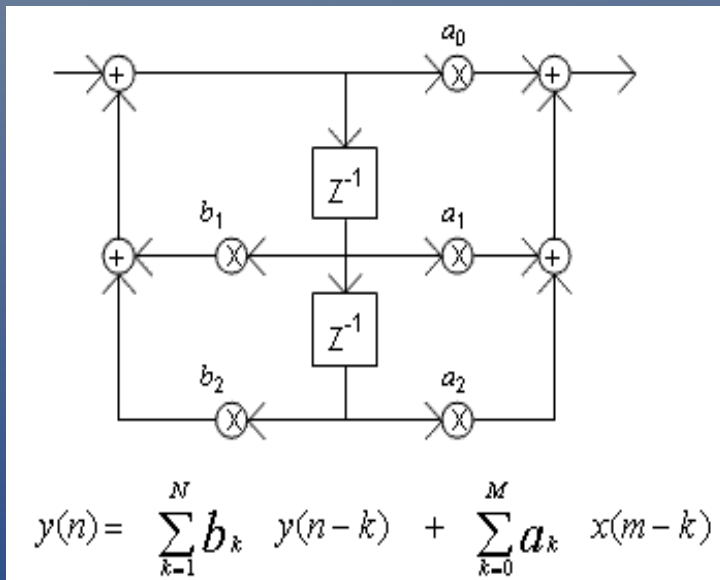
Control-feedback





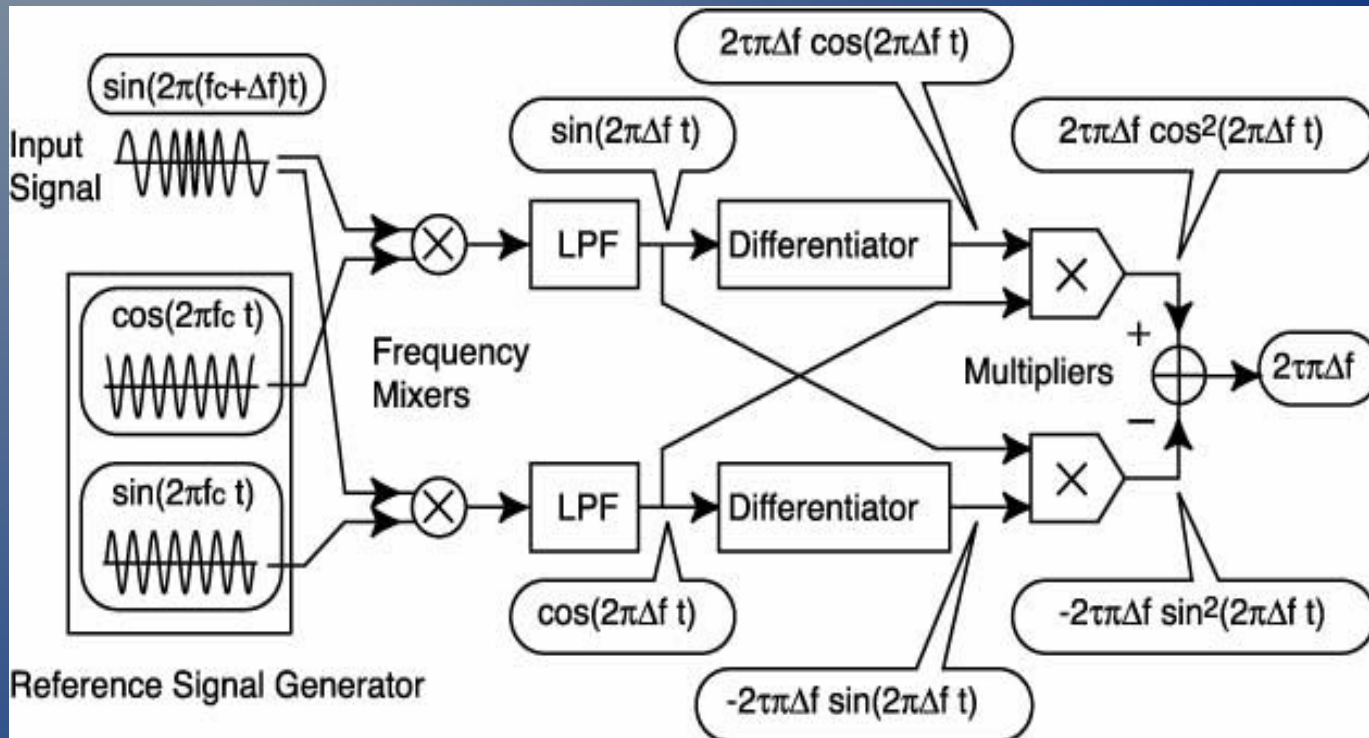
Infinite-Impulse (IIR)

Emulates RLC and Xtal filters well....low latency





Frequency Discriminator



Courtesy Kawakatsu Laboratories

$$\frac{Q \cdot dI - I \cdot dQ}{\sqrt{I^2 + Q^2}}$$



CORDIC Algorithm

COordinate Rotation Digital Computer

- Jack E. Volder, *The CORDIC Trigonometric Computing Technique*, IRE Transactions on Electronic Computers, September 1959
- Ray Andraka, *A Survey of CORDIC Algorithms for FPGA Based Computers*, FPGA '98. Proceedings of the 1998 ACM/SIGDA sixth international symposium on Field programmable gate arrays, Feb. 22-24, 1998, Monterey, CA. pp191-200.
- Iterative method for determining magnitude and phase angle
 - Avoids multiplication and division
- $N_{\text{bits}} + 1$ clock cycles per sample
- Can also be used for vectoring and linear functions (eg. $y = mx + b$)

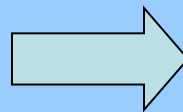


Concept

- Exploits the similarity between 45° , 22.5° , 11.125° , etc. and Arctan of 0.5, 0.25, 0.125, etc.
- Multiplies are reduced to shift-and-add operations

Angle	Tan ()	Nearest 2^{-N}	Atan ()
45	1.0	1	45
22.5	0.414	0.5	26.6
11.25	0.199	0.25	14.04
5.625	0.095	0.125	7.13
2.8125	0.049	0.0625	3.58
1.406125	0.0246	0.03125	1.79
0.703125	0.0123	0.01563	0.90

$$\begin{bmatrix} x' & y' \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$



$$\begin{aligned} x_{i+1} &= K_i [x_i - y_i \cdot d_i \cdot 2^{-i}] \\ y_{i+1} &= K_i [y_i + x_i \cdot d_i \cdot 2^{-i}] \end{aligned}$$



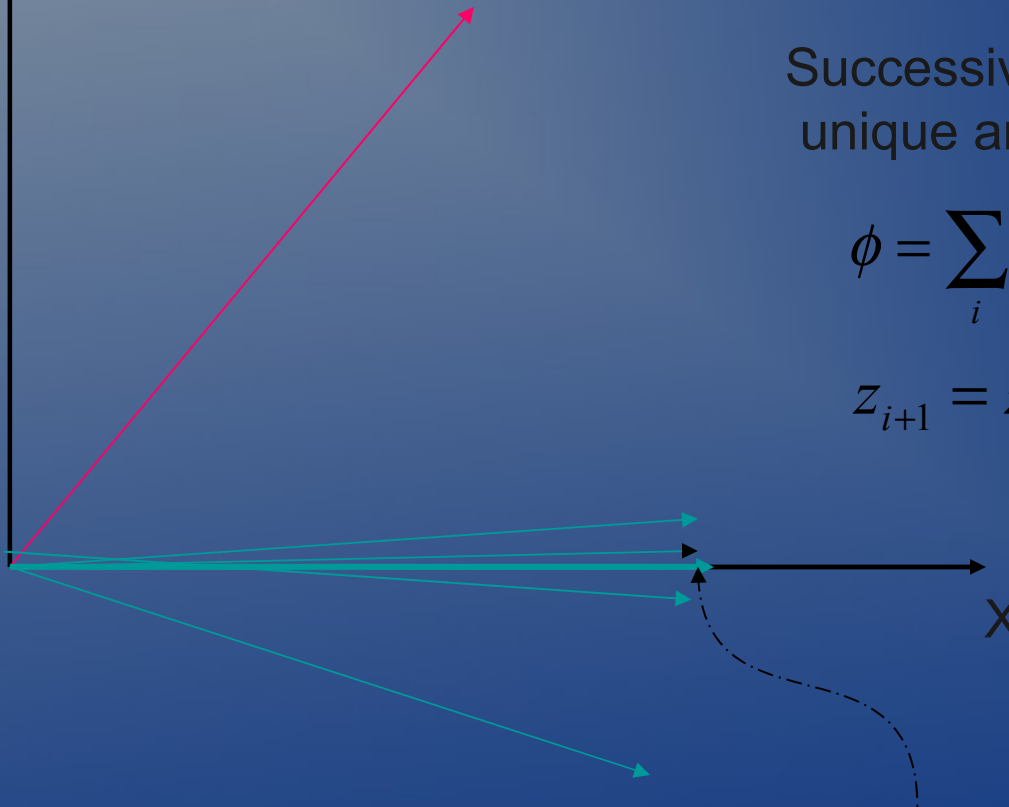
Functionally.....

Binary search, linked to $\text{sgn}(Y)$ $d_i = \begin{cases} +1, & \text{if } y_i < 0 \\ -1, & \text{if } y_i \geq 0 \end{cases}$

Successively add angles to produce
unique angle vector

$$\phi = \sum_i d_i \cdot \arctan(2^{-i})$$

$$z_{i+1} = z_i - d_i \cdot \arctan(2^{-i})$$



Resultant lies on X (real)
with a residual gain of 1.6

New Diagnostic Receiver Parameters

Low Noise Floor

Facilitate low beam currents

Ability to perform Y-factor noise calibrations

4-Channel design, for cavity and differential measurements

All digital, for re-configuration and pre-processing

Onboard IOC (PC-104) for lower cost, maximum flexibility

Relatively inexpensive, using consumer electronics

Accurate / repeatable AGC

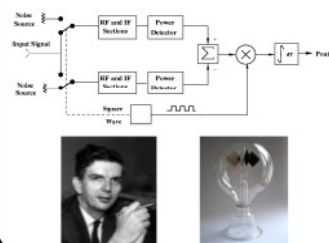
Reduction of Systematic Errors in Diagnostic Receivers Through the Use Of Balanced Dicke-Switching and Y-Factor Noise Calibrations*

J. Musson, T. Allison, R. Flood, J. Yan, Thomas Jefferson National Accelerator Facility, Newport News, VA

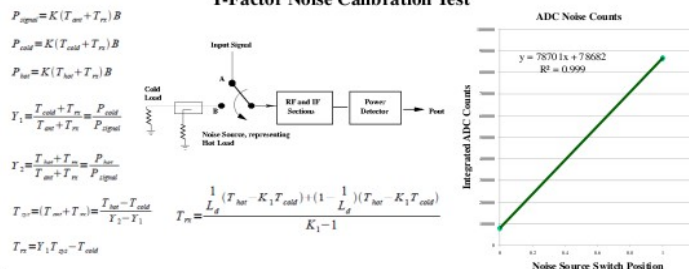
Abstract

Receivers designed for diagnostic applications range from those having moderate sensitivity to those possessing large dynamic range. Digital receivers have a dynamic range which is a function of the number of bits represented by the ADC and subsequent processing. If some of this range is sacrificed for extreme sensitivity, noise power can then be used to perform two-point load calibrations. Since load temperatures can be precisely determined, the receiver can be quickly and accurately characterized; minute changes in system gain can then be detected, and systematic errors corrected. In addition, using receiver pairs in a balanced approach to measure BPM X+, X-, Y+, and Y- electrodes reduces systematic offset errors from non-identical system gains. This paper describes and demonstrates a balanced BPM-style diagnostic receiver, employing Dicke-switching to establish and maintain real-time system calibration. Benefits of such a receiver include wide bandwidth, solid absolute accuracy, improved position accuracy, and phase-sensitive measurements. System description, static and dynamic modeling, and measurement data are presented.

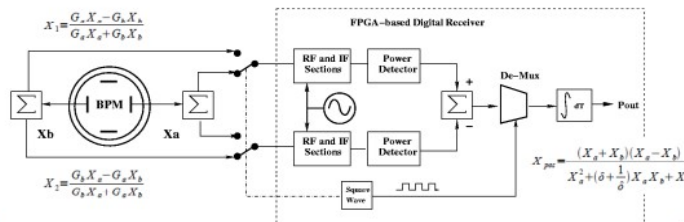
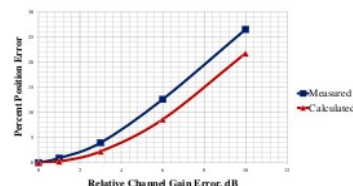
Dicke-Switched Radiometer



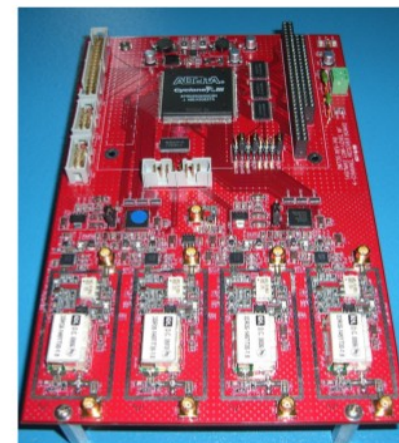
Y-Factor Noise Calibration Test



Balanced Dicke BPM Receiver Test



BPM/Diagnostic Receiver Board



Conclusions

Adaptation of a Graham/Dicke-switched receiver has been demonstrated. In addition, Y-factor measurements are easily incorporated into the switching process, to provide real-time calibration. The receiver must have a noise figure low enough to distinguish ambient noise. Commercial noise sources are available which can be applied to most systems with proper attenuation and coupling. On-board thermometry provides precise measurement of cold load temperatures, and can also be used to provide data correction through dead-reckoning. Finally, powerful FPGA platforms support extensive calculations and DSP algorithms, facilitating near-real time signal processing and self-calibration and correction.



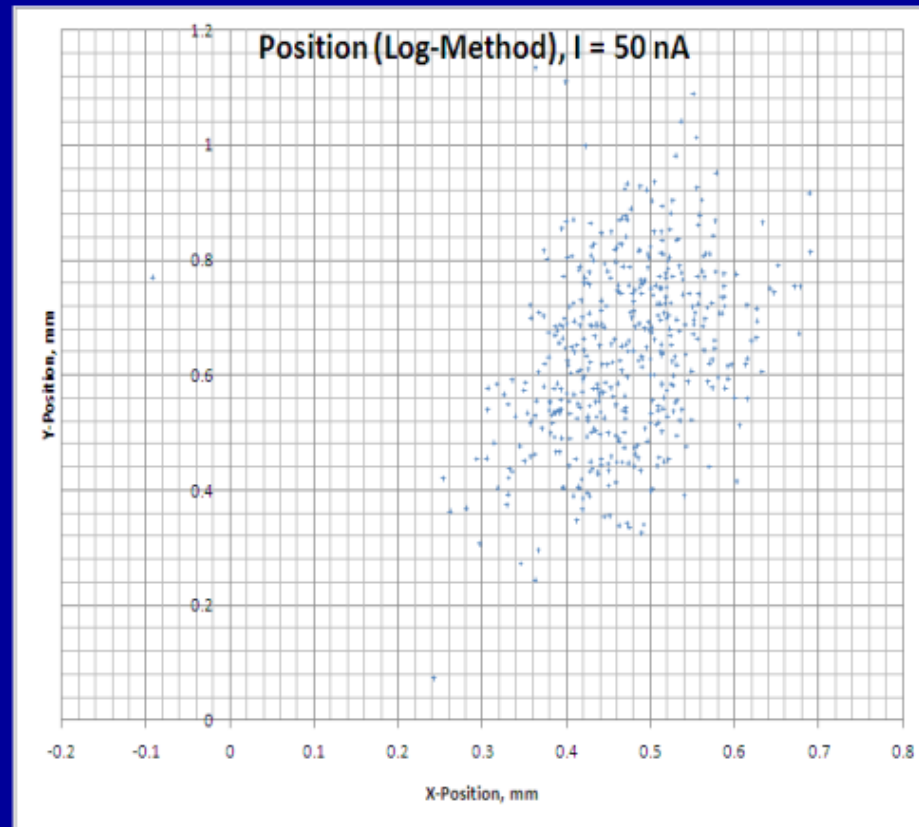
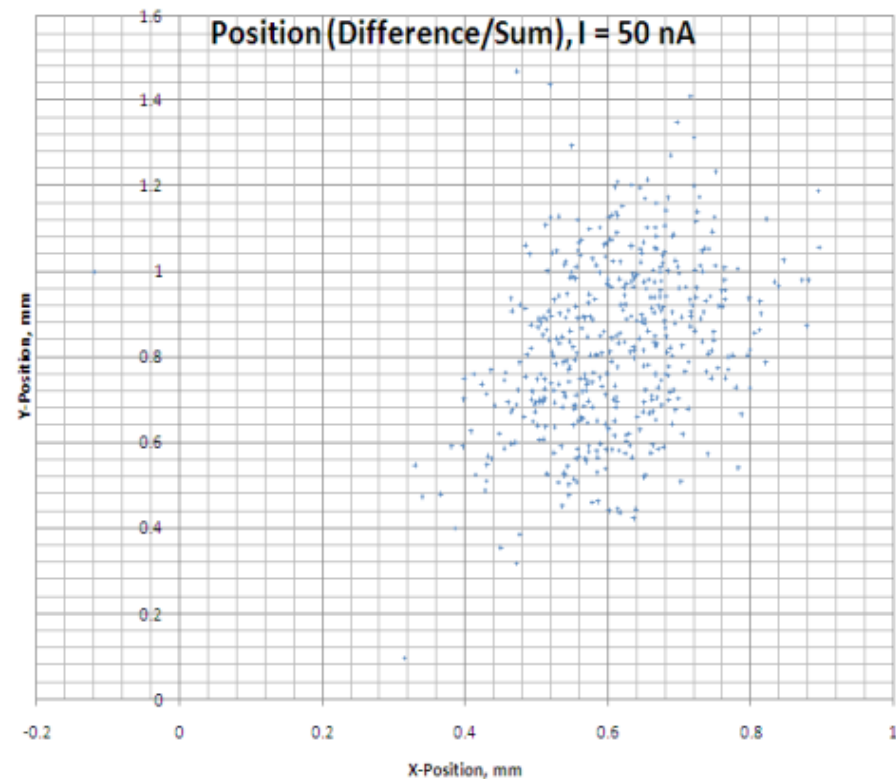
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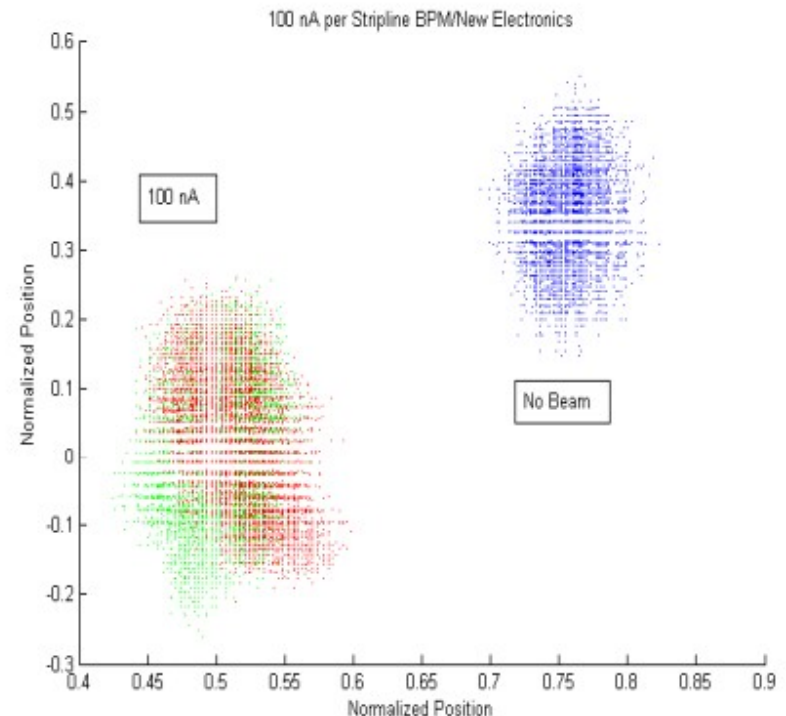
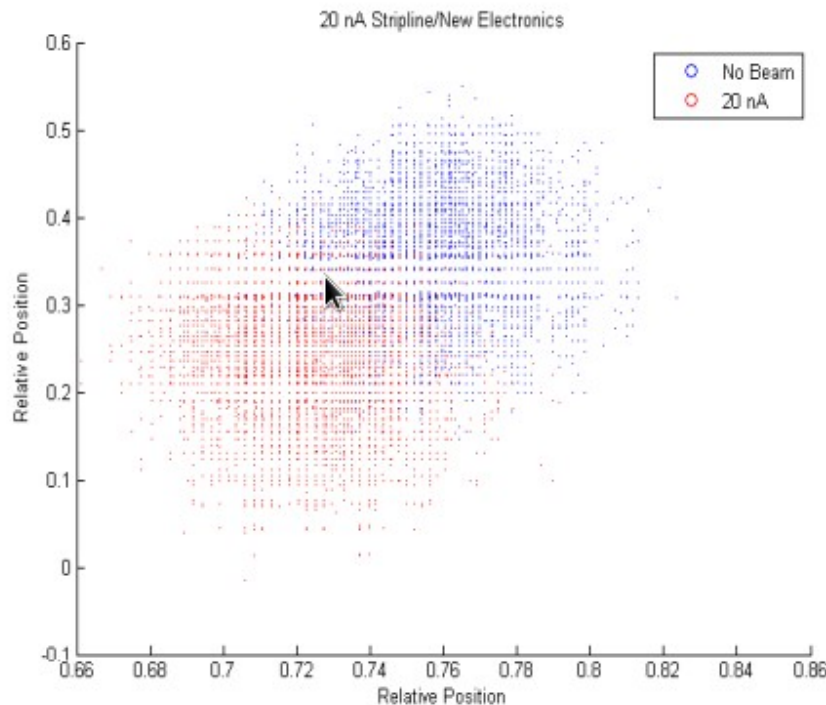
M15 w/ Prototype Electronics

5/11



System Test: Beam vs Wire

- Although beam is favored, it is often inconclusive
 - Beam motion and size can not be separated



Application Of Goubau Surface Wave Transmission Line For Improves Bench Testing Of Diagnostic Beamline Elements*

J. Munson, K. Cole, Thomas Jefferson National Accelerator Facility, Newport News, VA
S. Rubin, Rubytrom, Port Chester, NY

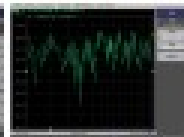
Abstract

In air test fixtures for beamline elements typically utilize an X-Y positioning stage, and a wire antenna excited by an RF source. In most cases, the antenna contains a standing wave, and is useful only for coarse alignment measurements in CW mode. A surface wave (SW) based transmission line permits RF energy to be launched on the wire, travel through the beamline component, and then be absorbed in a load. Since SW waves travel along the surface of the wire, the RF energy can be made to resemble the electron beam, limited only by ohmic losses and dispersion. Although many control systems are also a consideration, the diameter of the wire introduces large uncertainties in central location. A SW wire is easily constructed out of 200 micron magnet wire, which more accurately approximates the physical profile of the electron beam. Benefits of this test fixture include accurate field mapping, absolute calibration for given beam currents, 2-axis independence, and temporal response measurements of sub-nanosecond pulse structures. Descriptions of the surface wave launching technique, transmission line, and instrumentation are presented, along with measurement data.

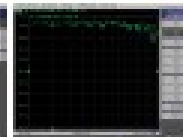
Goubau Line/BPM Test Fixture



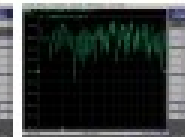
Insertion Loss (S11) plot of
1.6 mm diameter KoolWire



Return Loss (S11) plot of
1.6 mm diameter KoolWire



Insertion Loss (S11) plot of
300 micron diameter KoolWire



Return Loss (S11) plot of
300 micron diameter KoolWire

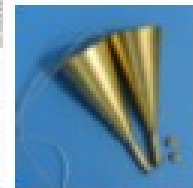
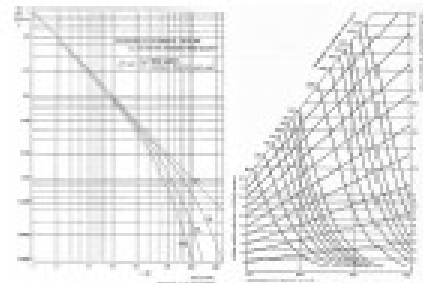
Goubau



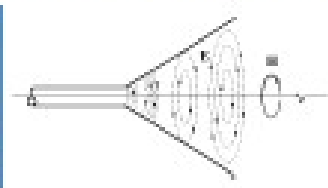
Conclusions

Traditional bench testing of beamline components will be inadequate to characterize and assess performance of the 12 GeV upgrade at Jefferson Lab. The use of the G-line facilitates measurements which more accurately mimic electron beam conditions. This system is particularly well suited for our bench system, due to ease of fabrication, low cost, and choice of operating frequency range. In addition, due to the flat 3 GHz frequency response, pulsed beam structures can be replicated, providing a platform for receiver development. Further reduction of VSWR is planned, in order to minimize dispersion of pulses resulting from reflections. Finally, the use of -1 cm X-Y stages presents a system which can be automated, improving repeatability and simplifying test procedures.

Development Of Surface Wave Launcher



30 Caliber Beam Prototype



Surface wave evolution inside the launcher

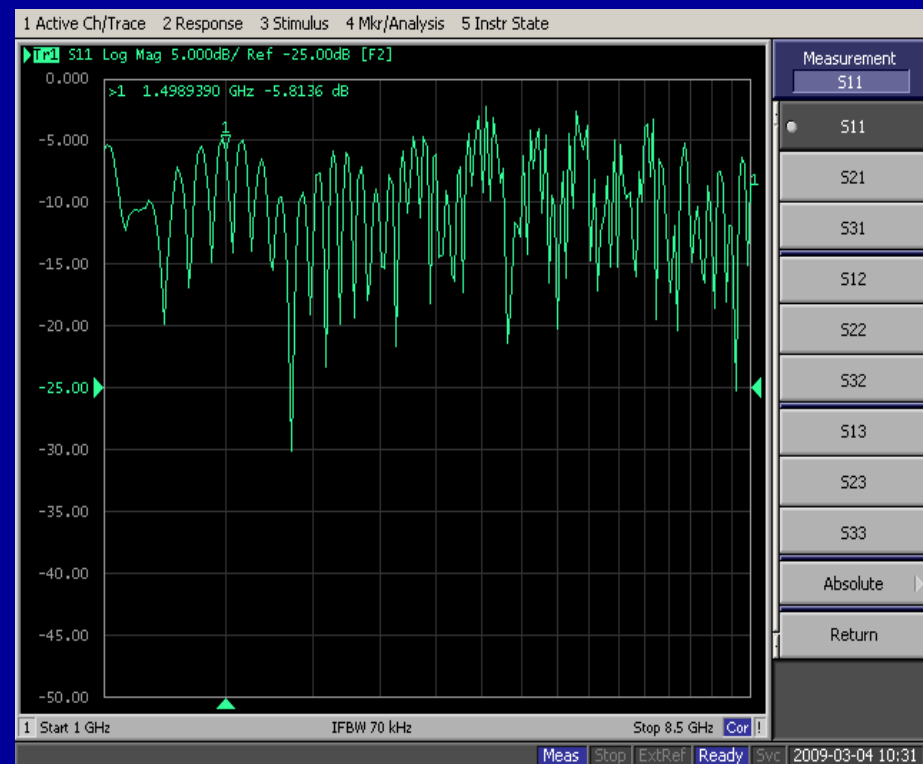
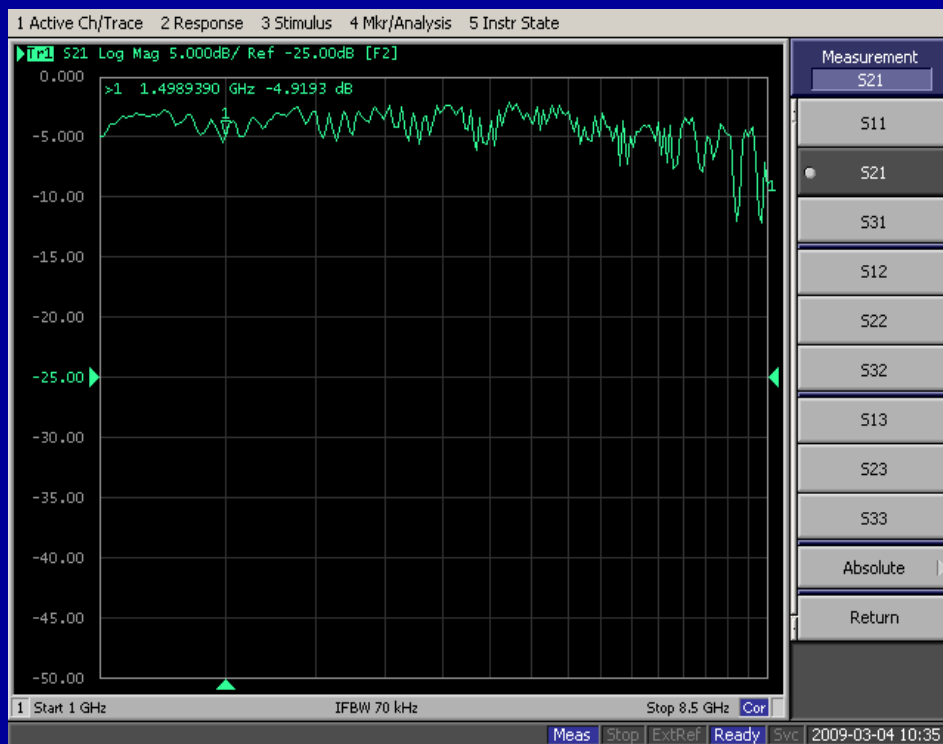
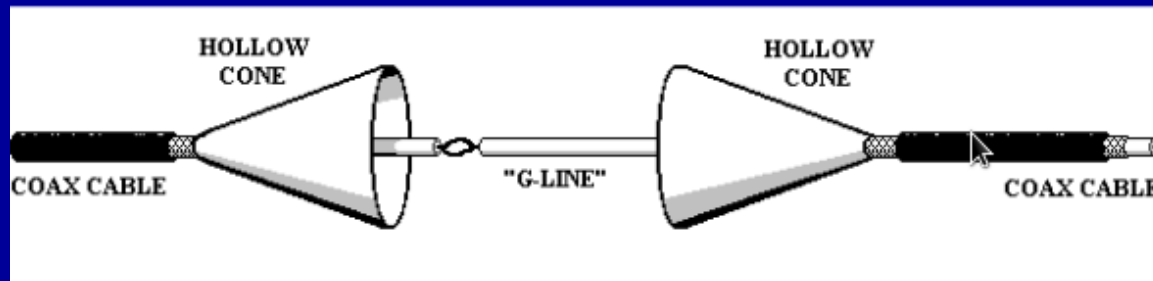


Commercial Rubytrom Inc.
KoolWire Launcher

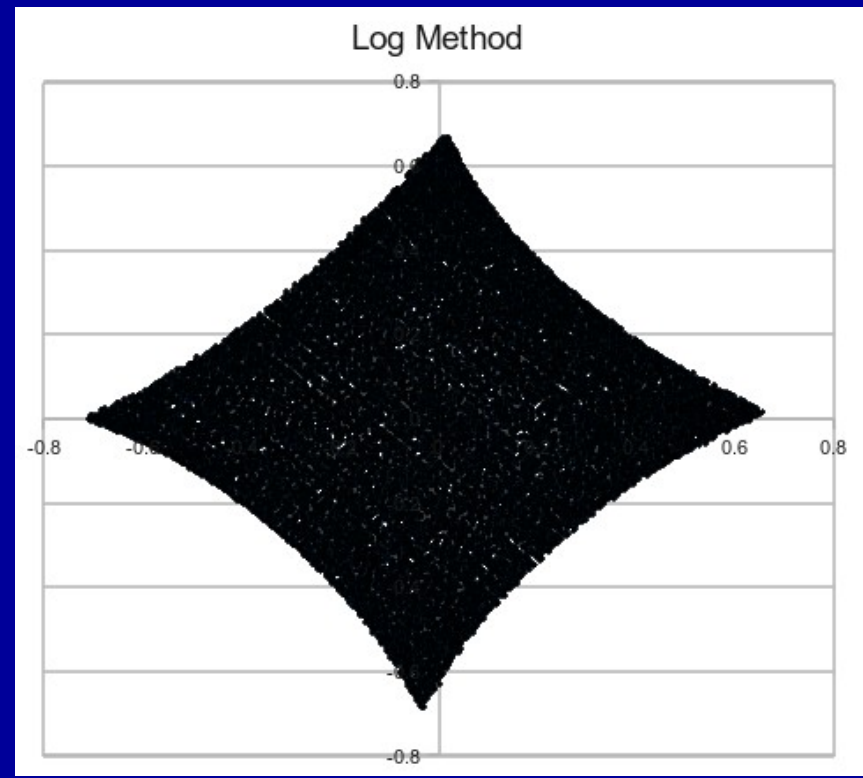
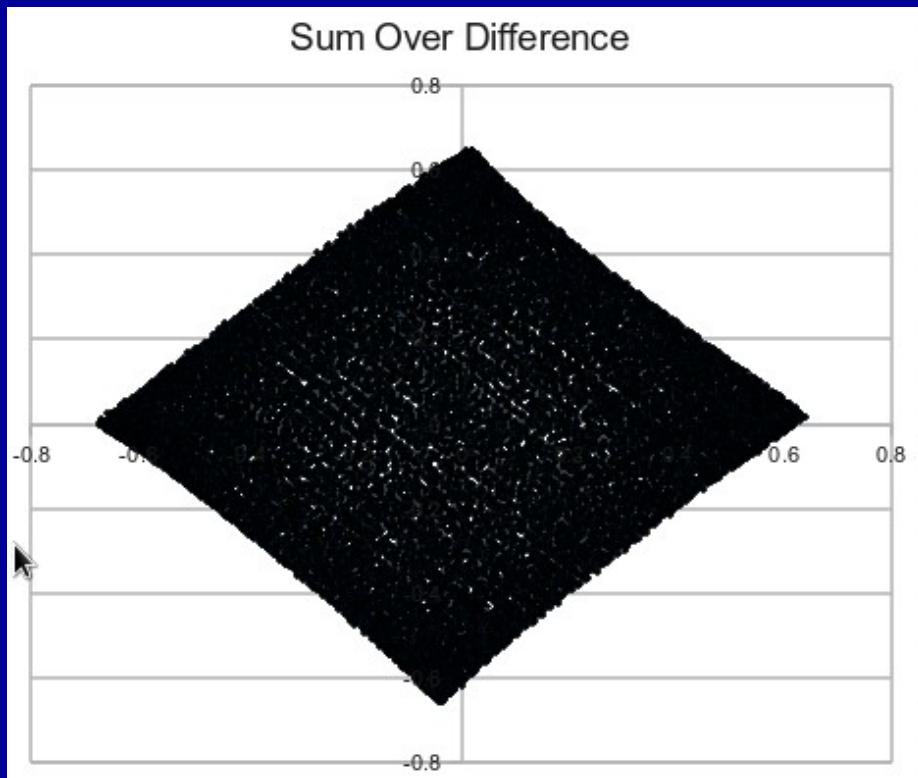


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G-Line Bench Tests of M15



ABSTRACT

The beam position monitors at the Continuous Electron Beam Accelerator Facility (CEBAF) have four antenna-style pickups that are used to measure the location of the beam. There is a strong nonlinear response when the beam is far from the electrical center of the device. In order to conduct beam experiments at large orbit excitation we need to correct for the nonlinearity. The correction algorithm is presented and compared to measurements from our stretched wire BPM test stand.

INTRODUCTION

Jefferson Lab's CEBAF Accelerator is a 1485 MHz 0.5 GeV protonated electron machine that has been providing high quality beam in excess since 1992. The accelerator beamlines are instrumented with over 300 4-wire antenna-style Beam Position Monitors.



Figure 1: 4-wire antenna-style beam position monitor

A linear relationship in determining beam position has historically been assumed for these devices with the X and Y positions calculated in the related form, according to:

$$X_{me} = k_x \frac{(V_1^+ - V_2^+) - (V_3^- - V_4^-)}{(V_1^+ - V_2^+) + (V_3^- - V_4^-)} \quad Y_{me} = k_y \frac{(V_1^+ - V_2^+) - (V_3^- - V_4^-)}{(V_1^+ - V_2^+) + (V_3^- - V_4^-)}$$

Where k_x and k_y are geometric scale factors that depend on the geometry of the BPM, V_1^+ and V_2^+ are calibration constants in terms of any difference in gain between a pair of antennas, V_3^- and V_4^- are the raw signal strengths from each antenna and the X and Y are the antenna signals with the beam and calibration signals off. This position of the beam in the XY plane is then calculated by a 40° rotation:

$$X = \frac{1}{\sqrt{2}} (X_{me} - Y_{me}) \quad Y = \frac{1}{\sqrt{2}} (X_{me} + Y_{me})$$

compared to the field problem of the antennas. The extent of the nonlinearity has been measured on a BPM test stand and also simulated using Poincaré.



CONCLUSION

The nonlinearity of 4-wire antenna-style BPMs has been modeled and measured with good agreement between simulation and test stand data. An algorithm was applied to correct the instrumental nonlinearity with better than 100 micron precision over a 750 mm grid of points.

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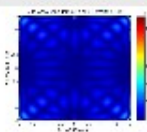
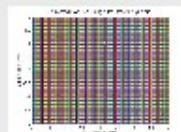
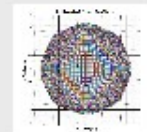
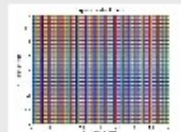
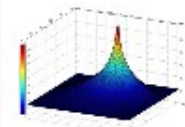
STRETCHED WIRE TEST STAND



- The electron beam is simulated with a P54 AWG magnet wire stretched between two RF cavities. The wire is 100 microns in diameter, which is comparable to typical beam sizes in CEBAF.
- A surface-wave SWR based transmission line probe RF energy is launched on the wire, travel through the beamline component, and then is absorbed in a load. Since SWR transmission lines employ traveling waves, the RF energy can be made to resemble the electron beam.
- The support stand is capable of moving the BPM in the horizontal and vertical plane in 10 micron steps.

CORRECTION OF BPM NONLINEARITY

- The BPM is modeled as a two-dimensional electrical problem using Poincaré. A potential map in one antenna is calculated and the symmetries of the system are used to generate maps in the other antennas.
- A square grid of points is used as a set of initial coordinates along with potentials calculated using the finite-difference method in potential x -two-dimensional polynomial fit.
- An 11th order polynomial removes the initial square grid of points in better than 100 microns.



*Submitted by Jeffrey Science Department, 11/2/2005 11:20 AM. Revised by Dr. J. Musson, 11/2/2005 11:20 AM. Revised by Dr. J. Musson, 11/2/2005 11:20 AM. Revised by Dr. J. Musson, 11/2/2005 11:20 AM. Revised by Dr. J. Musson, 11/2/2005 11:20 AM.

M15 Stripline (Evtushenko)

Conclusion

- Aggressively pursuing Beam Diagnostics Development
- MOU Draft underway with Bergoz
- New M15 to see battle, soon....
- Rx electronics to be used for '0L02, M55/M56, etc.



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